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IWRM
IN ISFAHAN

Integrated Water Resources Management in Isfahan, Iran
Zayandeh Rud Catchment Area

Sector Module 2: Urban Water Management
Subproject C: Value of Water

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0. KURZDARSTELLUNG ZUM SCHLUSSBERICHT

0.1 Das Verbundprojekt "IWRM Iran"

Das übergeordnete Ziel des Verbundvorhabens Integriertes Wasserressourcen-management in Isfahan, Iran "Zayandeh Rud Einzugsgebiet" (FKZ 02WM1177) war die Entwicklung eines wissenschaftlich fundierten, auf die spezifische politische, institutionelle, sozio-kulturelle und naturräumliche Situation abgestimmten und umsetzbaren, technologisch gestützten IWRM-Prozesses für eine nachhaltige Bewirtschaftung der Wasserressourcen in Isfahan, Iran. Ansatzpunkt des IWRM Prozesses war der enorme Handlungsdruck in der Region. Dabei zielte der IWRM-Prozess vornehmlich auf eine Integration der unterschiedlichen Sektoren bzw. Ressorts und auf die Entwicklung und Implementierung neuer Technologien und Management-instrumente ab. Das Verbundvorhaben zur Entwicklung einer integrierten und nachhaltigen Wasser-wirtschaft war in miteinander vernetzten Modulen strukturiert (siehe Abbildung 1) und fand in Kooperation mit der Wasserbehörde Isfahan statt.

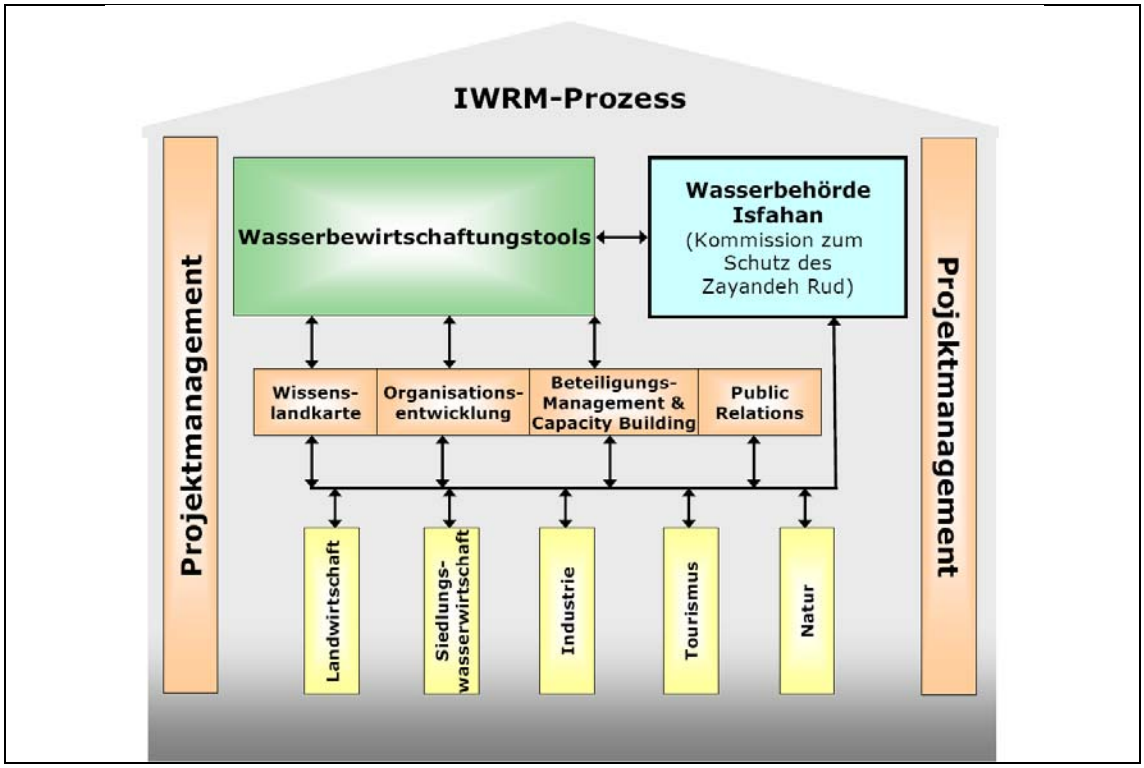


Abbildung 1: IWRM Isfahan Projektstruktur
[Quelle: www.iwrm-isfahan.com/de/project/structure.php]

Im Zentrum des Vorhabens stand, wie aus Abbildung 1 ersichtlich, das Hauptmodul "Wasserbewirtschaftungstools". Die fünf Sektormodule (Landwirtschaft, Siedlungs-

wasserwirtschaft, Industrie, Tourismus und Natur) berücksichtigten die Bedeutung und die Intensität der Einflussfaktoren aus den einzelnen Nutzungssektoren auf die Qualität und Quantität der Wasserressourcen im Untersuchungsgebiet.

Das Sektormodul 2 "Siedlungswasserwirtschaft" bestand aus 6 einzelnen Pilotprojekten:

(A) Verbrauchsmuster und Effizienz von Wassersparmaßnahmen,
(B) Reduktion von unbezahltem Wasser,
(C) Wert des Wassers,
(D) Abwasserreinigung zur Wiederverwendung in der Landwirtschaft,
(E) Einsatz innovativer Abwasserreinigungstechnologie,
(F) Klärschlammbeseitigungskonzept.

Pilotprojekt C ist Thema dieser Zusammenfassung.

0.2 Projektverlauf und Methodik

Zur Bestimmung des Werts des Wassers war die Erfassung und Analyse der technischen, organisatorischen, ökonomischen und ökologischen wasserwirtschaftlichen Ist-Situation im Einzugsgebiet erforderlich, bei der ein spezieller Fokus auf die separate Betrachtung der unterschiedlichen Wassernutzer gelegt wurde. Im Projektgebiet – das Zayandeh Rud Einzugsgebiet – ist der gleichnamige Zayandeh Rud Fluss von entscheidender Bedeutung für die Region. Der Fluss versorgt ca. 4,5 Millionen Anwohner mit Trinkwasser, den zweitgrößten Industriestandort im Iran mit Wasser für die Produktion und ist die Quelle für die Bewässerung von fast 260.000 Hektar Agrarfläche.

Die Lage im Projektgebiet ist von **akuter Wasserknappheit**, sinkender Wasserqualität und der drohenden Zerstörung des Ökosystems des nahegelegenen Salzsees Gavkhouni geprägt. Die Gründe dafür sind vielfältig. Neben klimatischen Veränderungen, einer über Jahrzehnte stark gewachsenen Bevölkerung, ökonomischer Entwicklung und der Ableitung von ungeklärten Abwässern, sind die bis dato meist politisch auf einen niedrigen Wert festgelegten Wassertarife, die **keine Allokation oder Einsparung des Rohstoffes Wasser** bewirken, ein bedeutender Faktor.
Aktuell führt die über Jahre **gewachsene Wassernachfrage** bei gleichzeitiger **Reduktion des Wasserangebotes** zu vermehrten Interessenkonflikten zwischen den

verschiedenen Abnehmern (Haushalte, Industrie und Landwirtschaft). Bisherige Versuche von iranischer Seite das Wasserangebot zu erhöhen, ohne die Wassernachfrage entsprechend zu steuern und zu verringern, waren nicht erfolgreich. Eine Anpassung der Wassernachfrage mit optimierter Wasserallokation und erhöhter Wassereffizienz ist erforderlich, um die Situation vor Ort zu verbessern. Die dafür notwendige Bestimmung des Werts des Wassers für die verschiedenen Nutzergruppen war Ziel des Pilotprojektes C.

Um den **Wert von Wasser** – und schließlich die Effizienz der Wassernutzung[1] – sektorübergreifend zu messen, war es im Rahmen dieses Projektes geplant, die Hauptwassernutzer (Landwirtschaft, Siedlungswasserwirtschaft und Industrie) für die ökonomische Evaluierung zu identifizieren und analysieren. Da der ökonomische Wert des verbrauchten Wassers nicht identisch mit dem gültigen und stark subventionierten Tarifpreis ist, sollte mit sogenannten **Schattenpreisen kalkuliert** werden, die anschließend in eine Kosten-Nutzen-Analyse eingehen können.

Der Schattenpreis lässt sich über eine sogenannte "Nachfragekurve" (= Kurve der Zahlungsbereitschaft) ermitteln. Dabei stellt das Integral unter der Kurve (punktierte Fläche, vgl. Abbildung 2) den Nutzen dar. Die Berechnung der während des Kalkulationszeitraumes ausgelieferten Wassermenge richtet sich nach der jeweiligen Wasserbilanz (resultiert aus Prognosen von Bedarf und vorhandenem Dargebot).

[1] Unter dem Gesichtspunkt, dass Wasser in Isfahan knapp ist, sollte es (aus makroökonomischer Sicht) dort eingesetzt werden, wo der zu erzielende Nutzen (Ertrag) maximiert wird. Bei dieser (Re-) Allokation sind nicht nur ökonomische sondern auch soziale und ökologische Kriterien mit einzubeziehen.

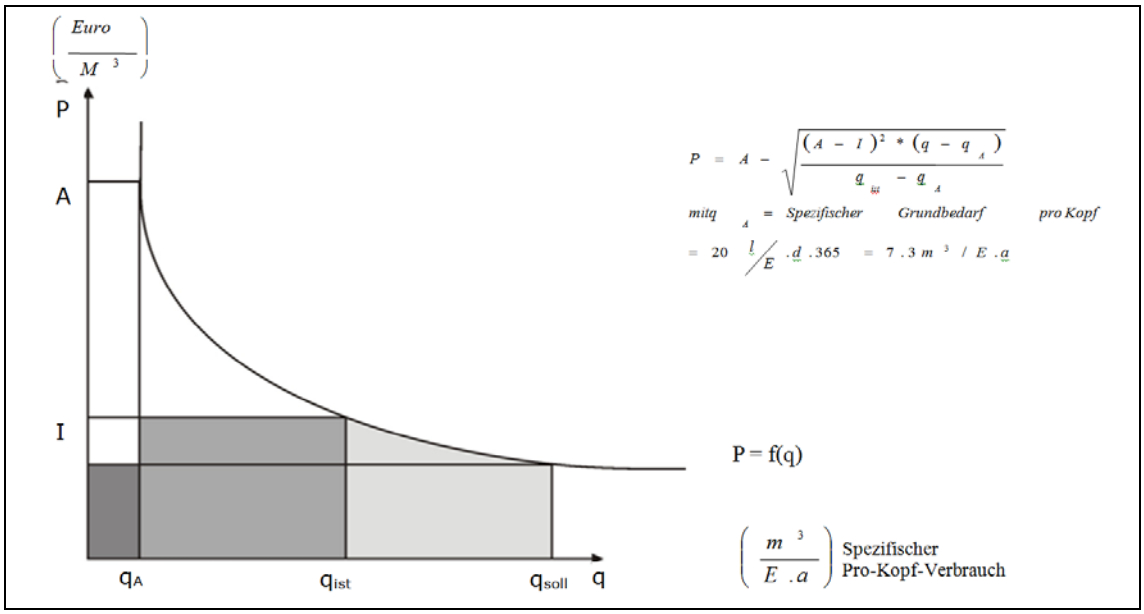


Abbildung 2: **"Nachfragekurve" (= Kurve der Zahlungsbereitschaft)**
[Quelle Rudolph, K.-U., 1980]

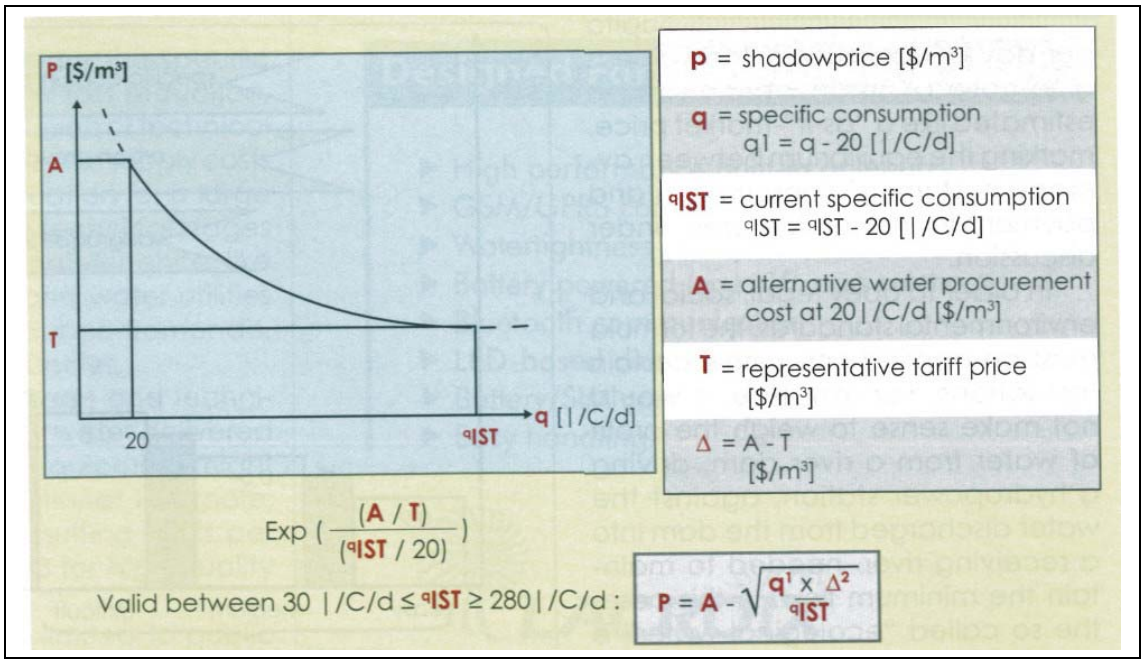


Abbildung 3: **Allgemeine Formel für die Validierung der Wasserversorgung**
[Quelle: Rudolph, K.-U., 2011, S. 9]

Um den Wert des Wassers mit Hilfe der Schattenpreise zu bestimmen wurden diverse Datensätze übersetzt und ausgewertet. Eine gezielte Auswertung und eine sinnvolle Interpretation der technischen, organisatorischen, ökonomischen und ökologischen wasserwirtschaftlichen Daten aus dem Pilotgebiet erwies sich jedoch als nicht möglich.

Der überwiegende Teil der übermittelten Daten war auf Farsi und musste übersetzt werden, viele Daten waren fehlerhaft und widersprüchlich und **die Datenbasis** dadurch für einen wissenschaftlich vertretbaren Standard **nicht ausreichend belastbar**.

Dies mag zum einem an der Tatsache liegen, dass der Iran eine "**Budgetökonomie**" ist und es kaum echte Preise gibt, welche am freien Markt entstanden sind und die Wertentscheidung der Käufer/Verkäufer für Güter und Leistungen belegen. Die Notwendigkeit zu optimieren und Handlungsalternativen abzuwägen (ähnlich dem Schumpeterschen Unternehmertypus (Schumpeter, 1912)) hat sich noch nicht durchgesetzt, wobei die Situation in den meisten anderen Transformationsländern ähnlich ist.

Desweiteren werden ökonomische Daten, sofern sie erhoben werden, nicht in der erforderlichen Form erfasst. Zum anderen setzte eine Weitergabe der Daten die Genehmigung etlicher Entscheidungsträger von iranischer Seite voraus, was wiederum zur Folge hatte, dass Datensätze gar nicht, verspätet oder unvollständig übermittelt wurden.

Zusätzlich zu der Datenproblematik sorgte die anhaltende Trockenheitsperiode und die damit verbundenen wasserwirtschaftlichen Probleme in der Projektregion für Verzögerungen, da die Projektpartner und Hauptverantwortlichen der lokalen Wasserunternehmen und -behörden sich auf Entwicklung von kurzfristigen Lösungsstrategien zur Reduzierung der Wasserkrise konzentrieren mussten und dieses Projekt nicht in dem angedachten Ausmaß unterstützen konnten.

Diese Umstände führten zu einer Verlängerung der Projektlaufzeit vom 31.08.2013 auf den 30.06.2014. Auch eine ex-post Komplementierung der Tarifdaten der Zielregion stellte sich als nicht durchführbar heraus.

Allerdings konnte das Forschungsteam stattdessen, in enger Absprache mit dem Projektträger und dem Projektkoordinator, eine Vergleichsanalyse durchführen, welche zum Ziel hatte, die Zahlungsbereitschaft der verschiedenen Nutzergruppen (und somit deren individuellen "Wert des Wassers") aus den Erfahrungen in anderen Ländern mit ähnlichen Rahmenbedingungen abzuleiten inklusive der Auswirkungen möglicher Tarifänderungen auf die einzelnen Nutzergruppen.

Dies geschah unter Berücksichtigung von drei für das Projekt entwickelten Szenarien. Somit konnte das Projektziel innerhalb des neu gesetzten Rahmens erfüllt werden.

0.3 Ergebnisse und Fazit

Grundsätzlich konnte im Rahmen des Projekts festgestellt werden, dass die Verhältnisse im Zielgebiet sich, wie erwartet, von denen in entwickelten Ländern (z. B. Deutschland) stark unterscheiden; sie weichen aber ebenfalls von anderen Entwicklungsländern ab, unter anderem dadurch, dass die öffentliche Infrastruktur nicht die kompletten Leistungen der Wasserversorgung umfasst. Abhängig von dem Einzelfall, der Nutzergruppe (Landwirtschaft, Industrie, Siedlungen) und den hydrologischen Verhältnissen (Versorgungsengpässe, Bedarfsdefizit permanent oder nur in Trockenperioden) hat sich eine Wasserinfrastruktur entwickelt, bei der die **Eigenleistungen der Nutzer** einen hohen Umfang ausmacht.

In der **Landwirtschaft** dominiert das System der Wasserverteilung über Bewässerungsgräben und Überflutungsbewässerung ("border dyke irrigation"). Dieses System ist robust und vergleichsweise kostengünstig; es wird aber ein Mehrfaches der Wassermengen benötigt, die bei effizienteren Systemen und Technologien verbraucht würden. De facto sind die überörtlichen Bewässerungsgräben überwiegend durch den öffentlichen Versorgungsträger installiert worden und werden durch diesen auch in Funktion gehalten. Ein anderer Teil, vor allem die Bewässerungsgräben "dritter Ordnung" zur Verteilung auf den landwirtschaftlichen Flächen oder zwischen einzelnen landwirtschaftlichen Flächen, wurde dagegen von unterschiedlich organisierten Interessengemeinschaften der Bauern oder einzelnen bäuerlichen Betrieben erstellt und werden durch diese auch bewirtschaftet.

Analog ist es so, dass in der Wassernutzergruppe **Industrie** die öffentlich finanzierte Wasserinfrastruktur nur dort die maßgebliche Versorgungsleistung stellt, wo beispielsweise Kleingewerbe oder städtische Industrie an das öffentliche Versorgungsnetz von geschlossenen Siedlungen angeschlossen sind. In den meisten Fällen wird die Industrie nur durch überörtliche Systeme versorgt und erbringt erhebliche Eigenleistungen durch Zwischenspeicherung, Nachaufbereitung usw. In sehr vielen Fällen versorgt sich die Industrie eigenständig.

Bei den **Siedlungen** wird der tägliche Wasserverbrauch in Städten der Provinz Isfahan auf ca. 250 Liter je an die Wasserversorgung angeschlossenen Einwohner pro Tag geschätzt. Zum Vergleich: nach Zahlen des Statistischen Bundesamtes beträgt der aktuelle Wasserverbrauch in Deutschland pro Person und Tag 121 Liter. Die Privathaushalte in Isfahan müssen im Gegensatz zur Landwirtschaft und Industrie wenig Eigenleistung erbringen.

Um sich dem Wert des Wassers für die Hauptnutzer zu nähern, wurden die Auswirkungen möglicher Preisänderungen auf das Konsumverhalten der einzelnen Nutzer analysiert, da dies den Wert des Wassers wiederspiegelt. Die Unterschiede und Problematiken die anhand der Analyse festgestellt werden konnten, müssen berücksichtigt werden, wenn ein ökonomisch und soziopolitisch angemessener Tarif im Rahmen des IWRM festgelegt werden soll.

Für das Zayandeh Rud Einzugsgebiet können die **Ergebnisse** wie folgt zusammengefasst werden.

Im **Landwirtschaftssektor** sind die Auswirkungen von Tarifänderungen auf das Konsumverhalten sehr schwer zu bestimmen. Die Wassereffizienz ist gering, aber es besteht das Potential sie deutlich und relativ einfach mit der Einführung von verbesserten Bewässerungstechniken zu erhöhen. Die Studien, die im Abschlussbericht analysiert wurden, ergaben, dass höhere Wasserpreise und Subventionen einen ausreichenden Anreiz für die Landwirtschaft im Iran darstellen, wassersparende Technologien einzusetzen. Eine erfolgreiche Umstellung der Bewässerung wird wahrscheinlich eine Verbesserung der Ernteerträge, der Schattenpreise und des ROI für die Bauern mit sich bringen. Allerdings gibt es mehrere Herausforderungen, die zu bewältigen sind (z. B. die korrekte Erfassung der Grundwasserentnahmen; die Berechnung von präzisen und fairen Wasserpreisen; die Problematik, dass höhere Preise kurzfristig das Einkommen der Landwirte mindern und das Landwirte oftmals entweder nicht in der Lage sind oder nicht gewillt sind, neue Technologien einzuführen und/oder gewinnträchtigere Agrarprodukte anzubauen).

Der Wasserbedarf **industrieller Abnehmer** im Zayandeh Rud Einzugsgebiet hat in den letzten Jahrzehnten stark zugenommen. Die Wertschöpfung in der Industrie ist dabei deutlich höher als in der Landwirtschaft (siehe Tabelle 1), obwohl die Wassereffizienz auch in der Industrie noch deutlich steigerbar ist.

Eine Studie von Rajabi (2013) für den industriellen Sektor in Isfahan hat festgestellt, dass eine Erhöhung der Wasserpreise die Wassereffizienz in der Industrie deutlich erhöhen wird.

Die Nachfrage der **privaten Haushalte** ist – wenn auch im Vergleich zu Industrie und Landwirtschaft sehr niedrig – ausgesprochen hoch, wenn man den Verbrauch pro Tag pro Kopf mit Industrieländern vergleicht. Wassertarife für Haushalte sind ein entscheidender Aspekt des IWRM, da Wasser nicht nur ein wirtschaftliches, sondern auch ein soziales Gut ist. Die Auswirkungen einer Preiserhöhung für den Wasserverbrauch der privaten Haushalte sind ein Thema, dass nicht nur in der akademischen Welt seit vielen Jahren diskutiert wird. Die große Mehrheit der Studien in diesem Bereich kommt zu dem Fazit, dass Preissteigerungen den Verbrauch reduzieren und kostendeckende Tarife notwendig sind, um eine langfristig nachhaltige Wasserversorgung zu gewährleisten, die Tarife für die Bevölkerung aber gleichzeitig leistbar und fair sein müssen. Im Gegensatz zu dem früheren Paradigma der stark subventionierten Tarife (die Wassertarife im Iran decken z. B. nur rund 10% der tatsächlichen Kosten) gilt es heute zunehmend nicht die Tarife zu subventionieren, sondern den bedürftigen Teil der Bevölkerung.

Um den Wert des Wassers zu bestimmen, wurden schließlich die Durchschnittstarife (USD/m³) und der Wasserverbrauch (L/Capita/Tag) von 192 Städten aus 19 Ländern der Jahre 2008 bis 2012 gegenübergestellt. Danach wurde die Stadt und Region Isfahan mit 70 Städten detailliert verglichen und die Unterschiede analysiert. Südafrika ist dem Iran in Bezug auf das BIP (PPP) der letzten Jahrzehnte sehr ähnlich und die Stadt Pretoria erwies sich in Bezug auf verfügbare Wasserressourcen (beide Städte sind relativ aride), dass Einkommen per capita und die Wasserinfrastruktur als vergleichbar.

Der Wert des Wassers wurde auf dieser Basis für drei vom Projektpartner ISOE angedachte **Szenarien** bestimmt. Die Szenarien sind: 1. Das "Trendszenario" (der Status Quo wird in der Projektregion beibehalten). 2. Die "Begrenzte sektorübergreifende Zusammenarbeit" (mit teilweiser Umsetzung der erarbeiteten IWRM Maßnahmen und ersten Preiserhöhungen in der Provinz Isfahan) und 3. Das "Nachhaltigkeit" Szenario (alle IWRM Maßnahmen werden umgesetzt inklusive Einbeziehung der Provinz Gavkhouni). Die folgende **Tabelle 1** bietet eine Übersicht über die Ergebnisse für alle drei Szenarien.

Tabelle 1: Wert des Wassers für Siedlungen, Landwirtschaft und Industrie für die entwickelten Szenarien

Nutzergruppe	Niedrige Wassereffizienz (Trendszenario)	Mittlere Wassereffizienz (Begrenzte sektorübergreifende Zusammenarbeit)	Hohe Wassereffizienz (Nachhaltigkeits-szenario)
Siedlungen (Haushalte)	3,00 USD/m³	5,00 USD/m³	8,00 USD/m³
Landwirtschaft (für Bewässerung und Tierbestand)	0,20 USD/m³	0,40 USD/m³	1,20 USD/m³
Industrie	4,00 USD/m³	7,00 USD/m³	14,00 USD/m³

Die Ergebnisse zum Wert des Wassers können wie folgt zusammengefasst werden: Der Wert des Wassers für landwirtschaftliche Nutzung ist deutlich geringer als der Wert des Wassers für private Haushalte deren Wert wiederum unter dem der industriellen Nutzung von Wasser liegt. Die hier genannten Werte stellen den ungefähren Wert des Wassers für Ver- und Entsorgung dar. Jedoch ist für die Abnehmer Industrie und Landwirtschaft zu berücksichtigen, dass zum Teil erhebliche Eigenleistungen erbracht werden, die den Wert der Wasserversorgung entsprechend verringern. Details finden sich im Schlussbericht.

Trotz der im Rahmen des Projektes erzielten Erkenntnisse über den Wert des Wassers, die Auswirkungen der Preisgestaltung auf die Verbraucher sowie Voraussetzungen für eine nachhaltige Wasserverteilung in der Zielregion, wird die **Einführung** einer neuen **Tarifstruktur** und entsprechender Wasserallokation, die den tatsächlichen Wert des Wasser zumindest zu einem gewissen Grad widerspiegelt eine **große Herausforderung** und sollte auch als politischer Prozess gesehen werden. Dieses Ergebnis ist sehr bezeichnend durch die folgende Aussage von Nikouei und Ward wiedergegeben (2013, p. 43):

> *Translating our results into a just water rights administration system will be difficult in Iran or other developing countries with world views in rural areas different from policy analysts schooled in western analytical thinking. That is, policy insights gained by rigorous scientific analysis of objective data have no guarantee of being used to inform or support policy.*

Dieses Zitat lässt erkennen, wenngleich aus Sicht der Autoren harsch formuliert, dass man bei der Adaption von technologiebasierten Lösungen und ökonomischen Konzepten nicht auf der für Industrieländern bewährten Basis verharren darf, sondern gegebenenfalls neue Ansätze finden muss (die aus deutscher Sicht als sehr pragmatisch kritisiert werden können, zur Überbrückung der Lücken aber notwendig sind).

Bei **Beibehaltung der bisherigen Strukturen** und Wassertarife wird sich die Situation im Projektgebiet weiter verschlechtern und mittelfristig zu einer **Wasserkrise führen**, da die Wassernachfrage bei einem sich reduzierenden Wasserangebot kontinuierlich steigt.

1. INTRODUCTION

The Zayandeh Rud River is vitally important for the economic development of the semi-arid region of Isfahan. The river provides drinking water for approximately 4.5 million residents. Additionally, it is the irrigation source for approximately 260,000 ha of agricultural land and provides water for the second largest industrial area in Iran. Climatic changes, progressive population growth and an emerging economy increased total water demand in the last decades dramatically. The river is now overused - water scarcity, poor water quality and the imminent drying up of the important Gavkhouni swamp (a seasonal salt lake) are the consequences. The combined effects of increased water demand and reduced water supply have caused intensified competition among the different water user groups (i.e. agriculture, industry and domestic users) and the imminent danger of a river basin closure (Yousefi, 2010). A river basin closure is an anthropogenic process that occurs once the supply is no longer able to meet the increased demands for an extended period of time (Molle et al., 2010).

Despite significant efforts in the last decades to secure sufficient water supply in order to meet increasing demand by constructing numerous dams (increasing the supply of water) and modernization of water infrastructure (reduction of water losses) – in many regions of Iran (acute) water scarcity prevails. In addition, the annual precipitation in Iran is only around 250mm; much less than the average precipitation in Asia with around 732mm (Yousefi et al., 2010). This also applies to the selected project area Isfahan. The very low annual precipitation is one of the reasons why only increasing the supply side to meet increasing demands will not work in the long term. For several years, the project area has been suffering from drought, which has further exacerbated the problem of water scarcity. All sectors suffer from this situation, industry, agriculture, households, tourism as well as the environment – the Gavkhouni swamp, formerly a natural paradise, fell almost completely dry.

The combined water demand of all sectors is several times higher than the available water supply. It is therefore not possible to simultaneously fulfil the equity, efficiency and sustainability objectives. Therefore, water allocation should be based on an appropriately developed and weighted water tariff system that is designed according to the selected criteria concerning optimal water allocation.

1.1 The IWRM Project in Isfahan, Iran

Integrated Water Resource Management (IWRM) provides a decisive contribution to solving conflicts of interest between agriculture, industry, inhabitants and the environment. There are numerous definitions for IWRM. One of the most cited and accepted one is the definition by the Global Water Partnership (GWP) that will be used in this paper:

> *"IWRM is a process which promotes the coordinated development and management of water, land and related resources, on order to maximise the resultant economic and social welfare in an equitable manner without compromising the sustainability of vital ecosystems"* (GWP, 2000, p. 5)

The basic principles of IWRM are designed to:

- protect drinking water sustainability as a resource vital to human beings and the environment
- manage water and land resources in integration with other related resources
- integrate the participation of different user groups and decision makers systematically into the water management process
- bear in mind the central role of women in the provision, management and protection of water resources
- sustainably maximise the social and economic use of water under fair conditions

The process of IWRM is based on the understanding that the resource water is an integral component of the ecosystem in the form of a natural as well as a social and economic commodity. As stated by the 1992 United Nations Conference on Environment and Development in Rio de Janeiro (1992, p. 197): "Integrated water resources management is based on the perception of water as an integral part of the ecosystem, a natural resource, and a social and economic good." The acknowledgement of water as an economic good helped to slowly increase the acceptance of economic models and theories in the water sector. While this acceptance is necessary in order to improve the performance of the water sector in many parts of the world, economic principles are still often ignored and most water utilities do not recover all of their costs.

In Iran there is a high level of interest in IWRM on account of the enormous pressure to take action. Teheran, the capital city, is also the headquarters of the UNESCO 2002 founded Regional Center on Urban Water Management (RCUWM) and could play an important role in the future as a multiplier for central Asia.

The objective of the German-Iranian collaboration is to establish sustainable water usage along the Zayandeh Rud while taking into consideration the different competing requirements. The focus of these activities is to improve the living conditions of the Iranian population. This is to be achieved by selectively embedding proven technologies and management methods in the specific regional conditions as well as through the activation of sustainable water use.

In addition to the transfer of modern technologies and management instruments, the IWRM process strives to integrate the participation of different user groups from agriculture, industry, growing cities and relevant governmental departments. The IWRM process is designed to identify and provide a sustainable solution to the different requirements of user groups and possible conflicts of interest between farmers, companies, residents and the environment.

The project plan comprises the five sector modules: Agriculture, Urban Water Management, Industry, Tourism and Nature. These represent the different water uses in the catchment area of the Zayandeh Rud. The scientific results will be compiled in four other integration modules, visualised and integrated in-process into the discussion (see Figure 1).

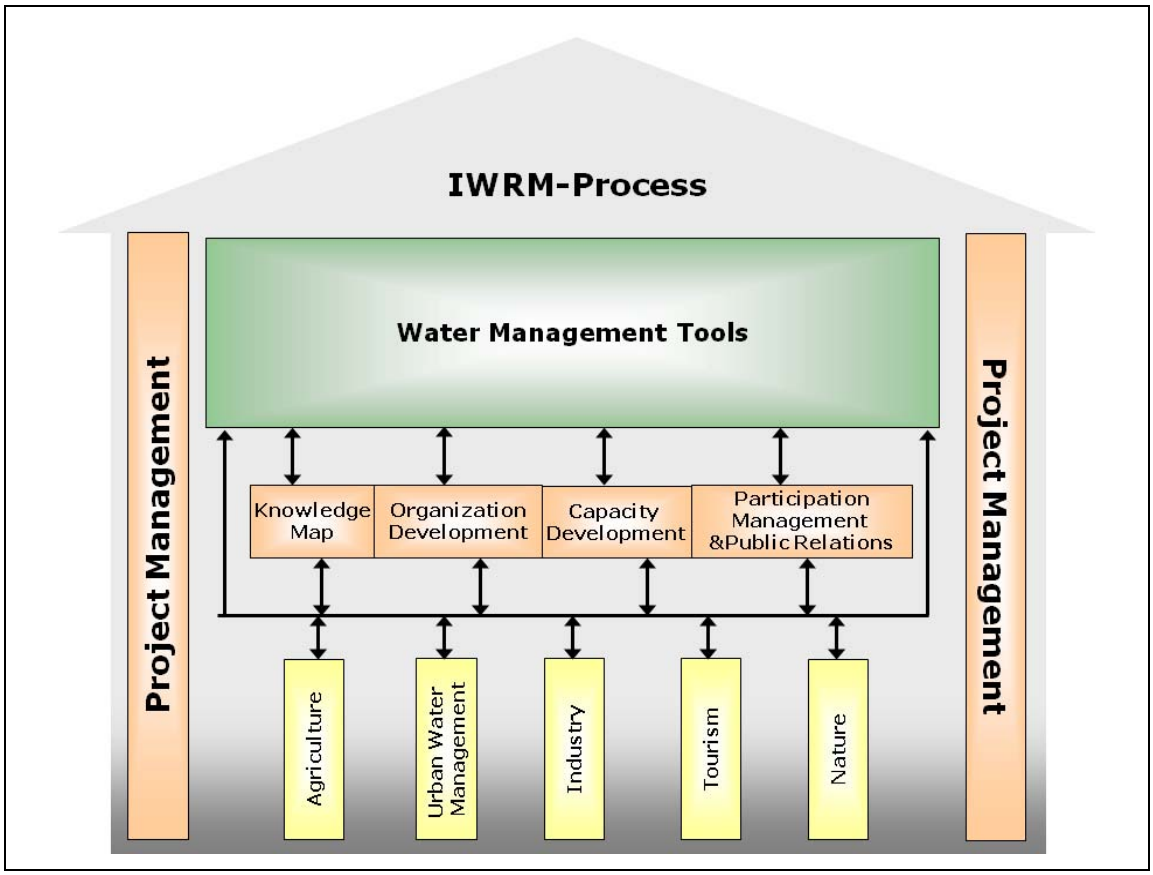


Figure 1: IWRM Isfahan Project Structure
[Source: *www.iwrm-isfahan.com/de/project/structure.php*]

The Sector Module 2 is "Urban Water Management" consists of six subprojects

(A) Efficient Water Consumption,

(B) Non Revenue Water Reduction,

(C) Water Value,

(D) Wastewater Purification for Irrigation,

(E) Innovative Wastewater Treatment Technologies and

(F) Sewage Sludge Disposal

Subproject C is the topic of this report. Its realisation will be presented in the next chapter.

Additional information about the other subjects and the IWRM approach can be found at: *http://www.iwrm-isfahan.com*.

2. PROJECT REALISATION

2.1 Contribution of the Project to the Funding Policy

The Framework Programme Research for Sustainable Development (FONA) of The Federal Ministry of Education and Research (BMBF) initiated the funding priority Integrated Water Resources Management (IWRM), in order to strengthen the concept of IWRM in developing and emerging countries, to support international cooperation in the water sector and to strengthen interdisciplinary cooperation of science, government and business. In addition, it serves to enable German companies in the water and environment sector to access and facilitate new markets. The joint research project "Integrated Water Resources Management in Isfahan" in Iran is part of the above funding priority, the research objectives were included in the project work plan, and the project thus contributes to the funding goals of the funding priority IWRM.

2.2 Requirements, Planning and Project Development

The idea of the project originated after the publication of "Long-Term Development Strategies for Iran´s Water Resources" by the Iranian government that showed not only a clear interest in IWRM but offered comprehensive strategies as well. In addition, IEEM gGmbH successfully carried out the BMBF funded project "Adaptation of water loss reduction programmes to the specific conditions of cities in Iran, illustrated by the City of Esfahan" (Reference No. 02WT0354) from 2002 to 2004 and already gained a lot of experience working in Iran.

German businesses have managed to maintain good commercial relationships with Iranian enterprises in other sectors for decades and Iran has then as now offered numerous very promising business opportunities for the water management sector. The capital expenditure for water utilities and equipment is projected to rise within the next years (GWI, 2014; see Attachment 9). Research in this area is necessary to improve the cooperation further, especially for the water management sector as the implementation of successful IWRM in the project region (and elsewhere in Iran) that suffers from acute water stress is urgently needed.

This alignment with the FONA IWRM objectives, combined with the strengthened interest in IWRM from the Iranian side and the extensive experience of the project coordinator inter 3 and the project partners with the Iranian market, led to the this project.

The notification of appropriation of funds from the BMBF for the subproject C of the main project (Integrated Water Resources Management in Isfahan, Iran) was sent by the Project Management Agency Karlsruhe Institute of Technology (PTKA-WTE) to the recipient of fund on the 07.09.2010. The project with the reference number 02WM1181 started on the 01.09.2010 and was planned to be completed on the 31.08.2013.

Compared to the original schedule, there was a slight delay mainly caused by the following two factors: Firstly, the severe drought in the project area caused many delays as the authorities were fully engaged in finding short term solutions for the acute water crisis and hence were not able to support this project to the same extent as intended.

Secondly, the figures and data that were provided by the Iranian side contradicted each other frequently, resulting in a situation where a scientific evaluation and a meaningful interpretation of data was significantly more difficult than usual or in many cases not possible at all.

On 27.11.2012, a corresponding extension request has been approved. The project was extended until 30.06.2014 and completed within this timeframe. The work focus of this Subproject was in the empirical area. Experimental work was not included.

2.3 Identification of current Scientific and Technical Knowledge

Due to the ongoing study and review of literature for the duration of this research project the current state of science and technology could be identified. The literature review has been conducted based on research material in the possession of IEEM, journal articles and relevant online directories of universities other libraries as well as specific specialized databases (e.g. "DVGW –Fachaufsatzdatenbank", DWA database), and the relevant databases according to the attachment" "Übersicht über Fachinformationszentren und überregionale Informationseinrichtungen" of the notification of appropriation of fund from the BMBF.

In addition, general online queries were conducted in which statements and conclusions of international institutions have been evaluated. Further discussions were held with experts from the private sector and at various conferences on IWRM.

2.4 Cooperation with other Parties

Participating research institutes and commercial partners are:

inter 3 – Institute for Resource Management (Project Coordinator)

- Core Module: Water Management Tools
- Sector Module 1: Agriculture
- Sector Module 3: Industry
- Sector Module 5: Nature

Institute for Social-Ecological Research (ISOE)

- Core Module: Water Management Tools
- Sector Module 1: Agriculture
- Sector Module 4: Tourism

DHI-Wasy GmbH

- Core Module: Water Management Tools

p2m berlin GmbH

- Sector Module 2: Urban Water Management
- Subproject A of Sector Module 2: Consumption Patterns and Efficiency of Water Saving Measures
- Subproject B of Sector Module 2: Reduction of Unaccounted-for Water
- Subproject D of Sector Module 2: Sewage Treatment for Agricultural Effluent Reuse
- Subproject F of Sector Module 2: Sludge Management Concept

Passavant & Watec GmbH

- Subproject E of Sector Module 2: Implementation of innovative Wastewater Treatment Technologies

In this final report, the results of the project partner IEEM gGmbH - Institute of Environmental Engineering and Management at the Witten/Herdecke University (Subproject C of Sector Module 2) are presented.

2.5 Use of Funds and obtained Results

The funds were used to achieve the following overarching goals

> To conduct research on the value of water in the Zayandeh Rud Basin including the City of Isfahan.

The specific objectives, challenges, the chosen approach as well as the final results can be found in the following chapters of this report.

2.6 Principal items of the financial statement ("zahlenmäßiger Nachweis")

The focal point of Subproject C 'The Value of Water' was on empirical studies. The corresponding tasks required a high number of personnel. Therefore, the staff costs presented the largest portion of the budget by far.

2.7 Necessity and Appropriateness of the Work Approach

As part of the research collaboration project Integrated Water Resources Management in Isfahan, Iran this subproject contributed to the implementation of the BMBF funding policy "Action concept for a sustainable, competitive German water-resources-management sector". The goal of the action concept is to design sustainable water supply and treatment concepts, particularly in growing and developing regions in the world.

This research project was based on the objectives of this action concept. It (as part of the research collaboration project) supports the international positioning of the German water management sector and to adjust and optimize the urban water management process technology used in Germany for a sustainable improvement of water management situations in other countries.

2.8 Utilisation of the Results

The benefits and the usability of the project arise from the contacts established with water experts in Iran, Germany and other counters as well as the experiences that were gained during the duration of this project. While it is understood that there are no blue-print solutions for the implementation of IWRM the challenges that were faced will provide an approach of how to overcome similar ones in the future, particularly in countries such as Iran. The approach chosen to determine the value of water for various scenarios might be of great help for future projects.

The work carried out within the framework of this project can be further utilised in order to realize the benefits that are to be expected once the ideas of the project are executed with the support of the contacts made and the collaboration with the partners of this IWRM project.

2.9 Progress in the Area of the Project Research in other Areas

The recipients of appropriations are not aware of any relevant progress made in this specific field of research during the execution of this project. However, numerous journal articles on the Iranian water market as well as the Zayandeh Rud Catchment area were published and analysed and taken into account by the authors (e.g. Yousefi et al., 2010; Rajabi, 2012; Nikouei et al., 2012; Nikouei and Ward, 2013).

3. SUBPROJECT C – AIM AND ECONOMIC ASPECTS

The objective of Subproject C is to conduct research on the economic value of water in the Zayandeh Rud Catchment area. The determination of the value of water for different uses is essential for many decisions within integrated water resources management (IWRM), such as whether to spend a certain amount of money for certain measures to improve water supply and sanitation, or to tolerate water shortages, i. e. supply below the demand of customers from industry, agriculture and domestic consumers, settlements.

The research of Subproject C is aimed to create a discussion and reflection of IWRM decisions based on economic considerations. This would be the starting point for dynamic development of water technology based solutions, such as water reuse, efficient water irrigation and the optimised allocation of water between different consumer groups.

Water and wastewater tariffs are key issues regarding the future balancing of demand and supply with water and wastewater services in the Zayandeh Rud Basin including the City of Isfahan, Iran. In the Zayandeh Rud Basin as elsewhere water demand is greater than water supply and since the water price is not based on the free market principle water is cheap, heavily subsidised and as a consequence being wasted. Water tariffs are well be below the level, which might influence the behaviour of water consumers and set incentives for saving water. Anyhow, a level of "price elasticity" to some degree higher than 0 (in order for prices to influence consumption significantly) would very likely be reached, once the water tariffs are increased to recover operational and maintenance costs or even full costs (including capital costs). The concept of water supply and demand, price elasticity and its impact on consumption will be explained in the following subchapters.

3.1 Water Demand and Price Elasticity

As is true for most traded goods, water consumption is negatively correlated to water prices (see for example Zetland and Gasson, 2013; EEA, 2013). This relationship is illustrated in the exemplary water demand curve in Figure 2; the curve can be interpreted as that of a single user as well as an aggregated demand curve. Each water user has its own demand curve, which reflects the price at which he consumes a certain

quantity of water. The slope of this demand curve expresses the price elasticity. It defines the relationship between demand and price of the good (the commodity here is not the water resource itself, but the final product which is largely determined by the water services provider) and thus the change in demand as a result of price variations.

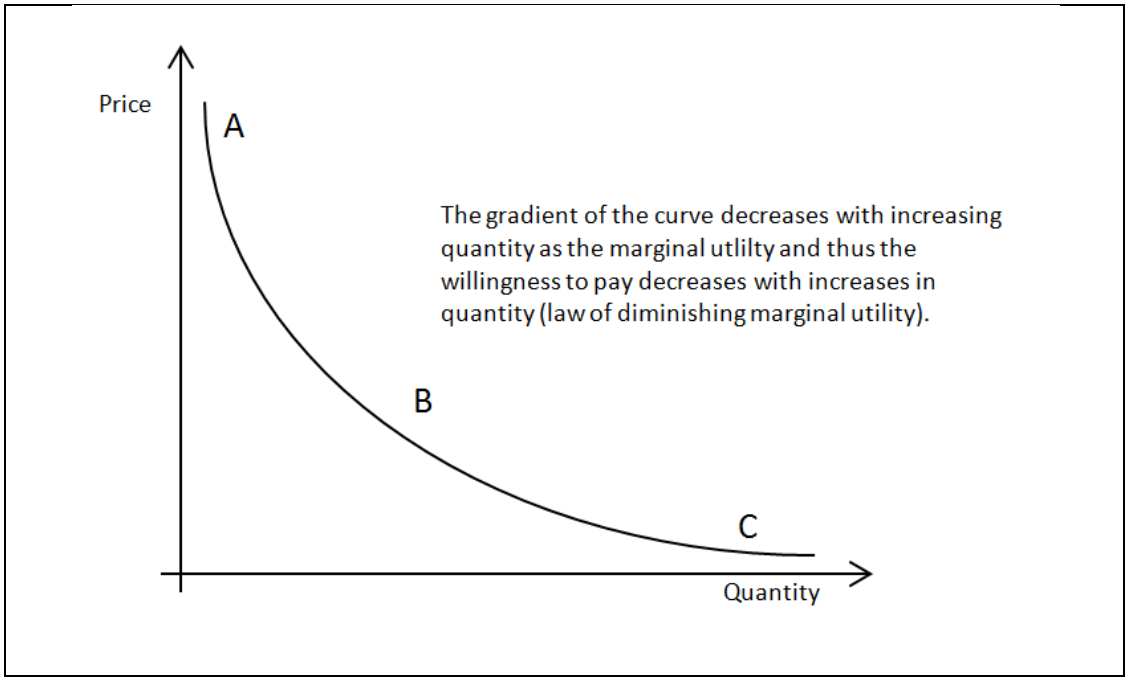


Figure 2: Price Elasticity of Demand
[Source: Rudolph, 1980]

The points A to C in Figure 2 can be described as follows:

Point A: The demand curve is parallel to the y-axis. The price elasticity of demand (PE) at this point is called perfectly inelastic (PE = 0). The water user consumes a certain amount of water regardless of an increase in price (the amount equals the distance between y-axis and demand curve) in order to satisfy his basic water requirements (e.g. drinking water).

Point B: At this point the slope of the demand curve is at 45 degrees. The demand is unitary elastic (PE = 1). A price increase of 1% causes a reduction in demand by 1%.

Point C: The demand curve is parallel to the x-axis. The price elasticity of demand at this point is referred to as perfectly elastic (PE = ∞). Perfectly elastic means that even at the lowest price increases, there is no more demand. In the example of Figure 2 is a convex demand curve, i.e. the price elasticity of demand increases with increasing

amount. Starting at point C the price is perfectly elastic (PE = ∞). Based on the price-quantity combination of point C, however, an increase in prices, unlike a linear demand curve with PE = ∞, does not mean a standstill of the total demand. The demand will only go back as far until a point is reached on the curve where PE < ∞ applies.

As illustrated in Figure 2 the willingness to pay for an additional unit of water decreases with increasing water consumption, since the marginal utility (i.e. the additional benefit of an additional unit of water) decreases. The decrease of the willingness to pay reflects the reduced value of water at higher quantities. The reason for this is that the water is – at least in theory – first used where it creates the most value. Looking at the water needs of any household, this is normally true for the coverage of drinking water needs. Only when this essential requirement is satisfied, water is used for other purposes. The same applies to water as a factor of production. It is first used for the production of goods that generate the highest marginal return and thus creates the most value.

All water users consume water until the marginal utility of one additional unit is less than or equal to the price for one additional unit (marginal costs). Thus, in regions with low (subsidized) prices water consumption is usually high and a lot of water is wasted.
In shortage situations, such as droughts, the willingness to pay due to the shortage of supply and the vital function of water (not only for the population but also for the environment and economy) is very high; especially so as it is very difficult to substitute water, if at all possible (→ low price elasticity). When there is no water supply via pipeline networks (due to lack of infrastructure or insufficient water supply), water is often supplied by private vendors that charge very high water prices of sometimes more than 10 EUR / m³ (e.g. in slums water is often sold by private vendors in small bottles for exorbitant prices).

Water supply-demand imbalances are one or the greatest threats to a sustainable use of water (Zetland and Gasson, 2013) supply as well as demand management is a crucial part of IRWM.

3.2 Water Tariffs and Shadow Prices

Setting water tariffs in Iran is, however, a political decision, as in many other countries worldwide. It is a matter of balancing socio-political against ecologic and financial benefits. This balancing process may be supported by scientific research, analysis and

advice. Yet by its nature, the final decision is not solely a matter of science and research. In dry regions and in regions where water is scarce, the value of water is high and in the case of subsidized tariffs much higher than the price paid for water supply.

The price to be paid by any of the different consumers (domestic users, industry, agriculture) for water supply and sanitation services is related to the following steering functions:

(1) The consumer should be informed about the value of water and water services and incentivised to use water efficiently to save water as it is an increasingly scarce resource. Furthermore,

(2) the water price paid is revenue for water utilities and necessary to maintain and operate the water infrastructure in a sustainable manner.

When setting water tariffs, two issues are to be considered:

On one hand, the price of water respectively the tariff structure should be defined in a way that it is accepted by the consumers as fair, reasonable and affordable.

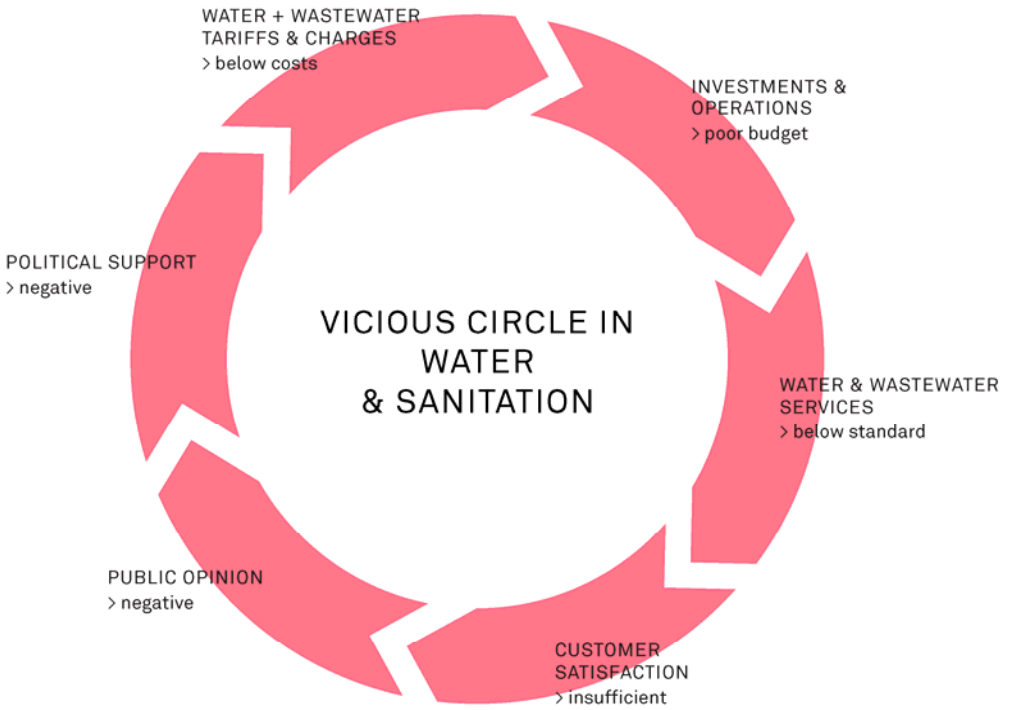
On the other hand, the tariff system must be organised in a way that it can be administered in day-to-day operations (i.e. not too complex, simplified, basing on parameters which are relatively simple to measure and easy to enforce).

There is a mutual interrelation between these two issues.

From a purely economic standpoint it would be ideal if water tariffs would cover all costs. However, in balancing economic and socio-political aims, a compromise often taken in the water sector worldwide is full coverage of running costs (expenses for operations and regular maintenance) excluding capital expenditures, i. e. for new investments, re-investments respectively structural maintenance.

No matter what kind of tariff is chosen: it is essential to know exactly about the economic efforts (i. e. the costs) which are generated on side of the utility to provide water services, year by year and cubic meter by cubic meter. In any sustainable IWRM system, all such costs must be fully covered by revenues from water tariffs or other financial sources (like permanent subsidies from government) - this is an essential pre-

condition for sustainable water management and operations. Whenever financial budgets are not sufficient to pay for extension, repairs and maintenance of existing water infrastructure, the water / wastewater utility will be forced to exploit its infrastructural and administrative facilities. The so called "vicious circle" in water and sanitation services "would be reality, instead of sustainable Integrated Water Resources Management" (see the following box).



In many developing and transformation countries, water tariffs are below costs. The utilities have to work with insufficient budgets. But, with an insufficient budget, investments and operations are below needs, which lead to poor water services, bad customer satisfaction and a negative public opinion. As a result, the political support (willingness to charge) to decide on cost-covering water tariffs, cannot be expected. This vicious circle could be broken, if all water produced would reach the paying customer. In other words: A proper water loss reduction programme is an essential pre-condition to reach sustainable water services.

Figure 3: The Vicious Circle of Water Services
[Source: Sewilam, H.; Rudolph, K-U, 2011; p. 115]

Therefore, the major aim of Subproject C is to quantify the value of water (which, in relation to the total costs expended, would allow to measure water efficiency in economic terms). Subproject C intends to evaluate water utilisation in the three main sectors of water consumption (i. e. agriculture, domestic use in settlements and industry).

Like in many other developing and transformation countries, the economic value of water served and used is different from the (subsidised) water price (respectively the water tariffs). Therefore, the classic calculation is based on shadow prices, one of the most important parameters for any cost-benefit-analysis (Rudolph, 1980; Nikouei et al., 2012). This shadow price for water can be generated as "demand function" basing on a so called "representative water tariff", which is the subsidised tariff plus the consumer profit (generated by the fact that water tariffs are subsidised or below market balanced prices, regardless of the reason).

The value of water, measured in quantity (assuming appropriate water quality, see above), depends on the specific use. For household consumption, a small part of the public supply might be used for drinking and nutrition. This is the most valuable fraction. Other valuable uses are sanitation and cleaning, whereas irrigation, car washing or wellness applications and swimming pools certainly have a lower value per cubic meter. For excessive consumption and water waste, the value of water will fall even further.

Therefore, the value of very little water available (for specific consumption) is extremely high, whereas the value for the last cubic meters of water consumption at a very high specific figure is extremely low.

This concept is visualised and calculated in Figure 4.

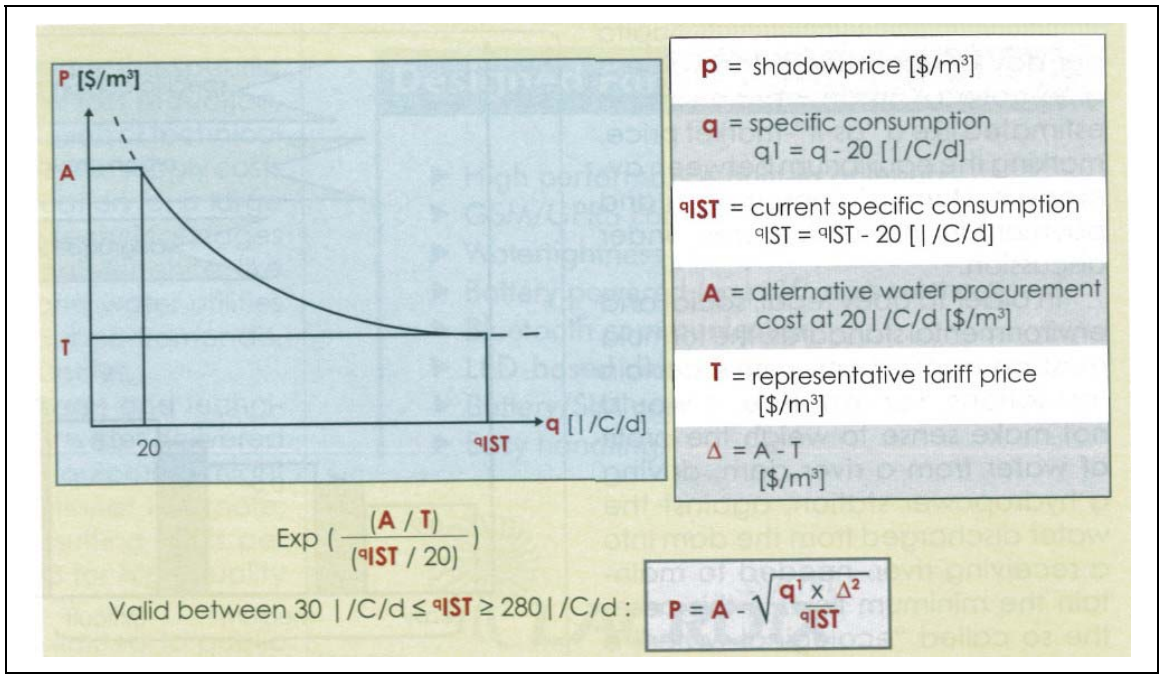


Figure 4: General Formular for the Validation of Supply Water
[Source: Rudolph, K-U, 2011, p. 9]

The formula is based on (at least one) representative point, for which a certain value (e.g. in Euros or Dollars per m³) can be fixed at a certain specific consumption rate (in litres per capita and day, or similar). The value is a so-called "shadow price", to be estimated like an "as-if"-market price, marking the equilibrium between average customer consumption and payment in the service area under discussion.

Note: To meet legal, social and environmental standards, the formula must only be applied within certain restrictions. For instance, it would not make sense to weigh the profit of water from a river dam that was driving a hydropower station, against the water discharged from the dam into a receiving river to maintain the minimum flow. In this case, the so called "ecological water" is excluded from any consideration, and its benefit would be equal to the minimum costs ("opportunity costs") of providing it.

Similar functions can be elaborated for other than domestic water consumption, e. g. for different branches of industry, etc. The more costly water is, the more the consumer (if charged for water consumption properly) will start to save water, even installing recycling systems, etc.

The analysis of the economic water value (to be elaborated with this Subproject C) is based on existing studies, collected through the Iranian project partners and the IWRM project coordinator. Technical input data will come from other sub-projects.

For the economic analysis, it must be understood and taken into account that Iran is a so called "budget economy" (mainly a state economy, not a full/free market economy). Many prices are not set by decision of consumers, not balanced between free production / supply and consumption / demand. Political decisions by the leaders of the country have been made for investments, organising operations in production and consumptions, in the water sector as in many other sectors. Therefore (as mentioned above) there are few prices which can be considered "real market prices", and any economic analysis will have to take other parameters like "shadow prices", "alternative cost calculations", "individual opportunity costs" etc. into account.

3.3 Water Allocation

Especially in areas of water shortages a sustainable and efficient allocation of water resources is essential as the supply is not sufficient to cover demand. Instead of solely focusing on increasing the water supply by building more dams, water desalination plants etc. the demand-side water management concept aims to increase efficiency and productivity of water used (see Palaniappan et al., 2006). The available water is preferably used where it provides the best economic efficiency (specific utility per cbm of water). If the same value can be generated with a smaller amount of water (or a larger value with the same amount), this means an increase in overall efficiency of the water used.

As a key element of IWRM, economic aspects in the allocation of water resources (Figure 5) are used to identify the water use efficiency of individual users and to identify opportunities for optimization in the water resource allocation and thus to increase economic welfare.

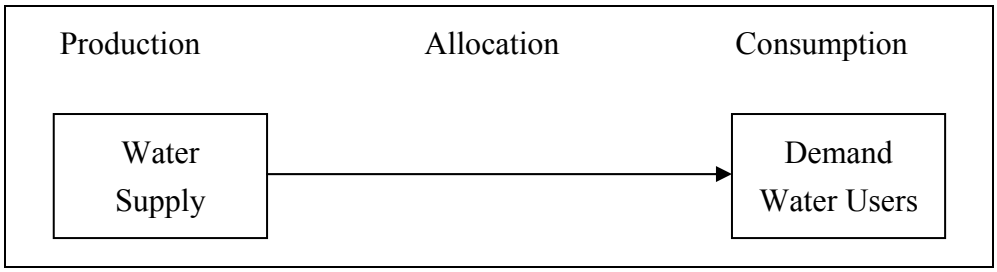


Figure 5: Water Allocation
[Source: Bombeck, 2013]

The economic goal of IWRM is to maximize the benefit that could be generated with the available water resources. For instance, the transfer of water from agriculture to high value urban or industrial uses – up to a certain extent – would increase the overall benefits (Srinivasan et al., 2013). If the total utility can no longer be increased by redistribution (reallocation) of the commodity, then – from an economic perspective – the optimal allocation of a good is achieved.

For the modelling of an optimal resource allocation, perfect market conditions are assumed in the sense of the theory of neoclassical economics (see, inter alia, Walter et al. 2010). In addition, several other simplifications are made (e.g. water quality, return flow, distribution and transport costs are often not integrated in these models). As a consequence, assuming a perfect market with a free market economy is problematical and will not create a realistic model for optimized water allocation for a specific region, particularly in a budget economy such as Iran.

Moreover, the occurrences of externalities, which are not integrated into the model, play a crucial role. External costs or benefits are experienced by parties that did not create the cause for the incurring costs or benefits. For example, external costs occur when the water use and contamination of one party decreases the water quality (i.e. the value of the water) for other users.

The results of economic models regarding total utility maximization may therefore – due to the necessary but not always realistic assumptions – not be used as the sole criterion for the pricing and allocation water resources. The results should rather only be a part of further (new institutional economics) aspects and supplemented in an integrative approach, with particular attention to social and environmental issues (Bombeck, 2013).

An allocation of water in a way that all social, environmental and economic objectives are fulfilled will not be possible with a demand that is larger than the (sustainable) supply. Therefore, the objective of IWRM is to find the right balance of a three objectives that is tailor made for the specific region and population.

4. THE ROLE OF DIFFERENT WATER TARIFFS WITHIN IWRM

In the previous chapters, the objective of this research project has been explained as well as the price elasticity of demand, water tariffs and shadow prices in order to determine the value of water. Before the value of water is determined it is essential to be aware of the different water tariffs that are used as well as their general and specific impacts on the consumers. This and the following chapter will provide background information necessary for a comprehensive understanding of water prices within the context of IWRM. So far the water and wastewater tariffs have only been taken as proof that customers are willing to pay a certain amount, and still continue to use water at this price level. The indicated value of supply water and wastewater services (based on real water tariffs plus an estimated consumer profit, resulting in the "shadow price" should not be misinterpreted: The value of water is no recommendation for setting-off water and / or wastewater tariffs.

In the context of IWRM, the water and wastewater tariffs play a crucial role. The higher tariffs are, the higher is the economic driver towards efficient water use, with less waste of water, more water reuse and (for example) efficient irrigation etc. (see for example Rudolph, 2014)

On the other hand, it is obvious that water and wastewater tariffs must be set in a way that does not overstretch the socio-political potentials, the ability to pay, willingness to pay and willingness to charge.

Finally, and from the economical / financial point of view, it is not reasonable to spend less for water and wastewater than indicated by the economic water value (or, vice versa: to implement measures with higher marginal costs than marginal benefits). For instance: the investments and operational expenses for the implementation of a water reuse scheme are only profitable, if the total value of water is exceeding the costs. But on the commercial level from the viewpoint of investors (e.g. industry, water utilities), water recycling schemes do not make sense financially, as long as the savings for water tariffs, energy, valuable substances etc. do not even cover the costs sufficiently. Saudi Arabia, as another example, has concluded that it is cheaper to reduce demand than to increase supply and is consequently decreasing its domestic wheat production in order to reduce water consumption substantially (Cotula et al., 2009).

One of the aims of charging for water on a volumetric basis is incentivizing recycling and reuse of wastewater as well as rainwater harvesting.

Under IWRM, it may be appropriate to work with subsidised water and wastewater tariffs (to protect the poor or else) as long as subsidies or other financial sources than water / wastewater tariffs recover the costs, including sustainable maintenance, at the optimum level.

In water scarce regions like the Zayandeh Rud Catchment, the water tariffs should, however, at least reflect consumption. Flat rates (without metering) are not considered to be reasonable in that context. Unfortunately, water metering is far from being adequate for households as of now.

As a guide for further considerations, some economic considerations for the selection and implementation of water and wastewater tariffs are provided below. The implementation of a suitable tariff system for a specific region that is accepted by the consumers is vital for the success of IWRM. This chapter focuses on general aspects of water tariffs that is necessary provide a basic understanding for the following chapters that will discuss more specific impacts of water value and prices within IWRM related to the target region.

4.1 Fixed and Working Charge

According to the Independent Pricing and Regulatory Tribunal (IPART) of New South Wales, a two-part tariff with a fixed charge (USD/month and meter or so) and a working charge (USD/m³) is the most efficient pricing structure for monopoly service providers. This is because the variable charge can be based on the marginal cost of supply while the fixed cost can recover the difference between the average cost and marginal cost (Cox, 2010). One example for a fixed and working charge is Germany.

Due to its federal system Germany does not have one central legislative framework for (waste-) water tariff calculation but each federal state and its municipalities have its own legislation that various slightly from the others. In general, water supply and wastewater treatment services can be offered by public or private organisations. Regardless of the organisational structure all water tariffs have to be set in line with the corresponding legislation.

For instance, the guideline issued by the Bundesverband der Energie- und Wasserwirtschaft e.V. (BDEW) offers a conclusive overview of tariff setting and calculation regulation and legislation for indirect wastewater.

All federal states have to implement wastewater tariffs that adhere to the

- benefit principle, i.e. an appropriate balance between the service offered and wastewater fees
- equality principle, i.e. arbitrary procedures and preferential treatment are prohibited
- cost recovery principle, i.e. the wastewater fees should cover all organisational costs, operating and maintenance expenses, depreciation and return on capital. The capital asset pricing-model is, for instance, recommended in order to determine the imputed interest rate
- prohibition of cost overrun principle, i.e. the wastewater utility is not allowed to charge more than it costs to offer and maintain its services

For economical reasons wastewater fees are calculated based on probability, i.e. utilization of services based on indicators like fresh water consumption not based on actual utilization.

There is an ongoing debate in Germany whether to increase the basic fee for water and wastewater as the capital expenditure for providing water services constitutes a large part of overall costs with up to 50 %, will remain the same in the medium/long run regardless of population fluctuation (BDEW, 2011). The debate is inter alia caused by continued urbanization and a decline in population that led to a declining wastewater flows and decreased demand for wastewater treatment and drinking water. Due to this development the old tariff structure is questionable and in need of replacement. An increased basic fee could cover the fixed costs and would prevent higher tariffs for water/wastewater consumption that would in turn lead to a decreased overall consumption – due to increased water saving efforts by the population – and hence decreased revenue. Nevertheless, in line with the European WFD Germany has achieved a very high cost recovery rate for water and wastewater services (BMU, 2011).

4.2 Progressive Tariffs

Increasing block-rate tariffs (IBT) are increasingly being adopted by utilities in developing countries (World Bank, 2010). With IBTs, the amount of water required for essential uses can be provided at a low price per unit, while consumers that use large volumes of water per month are charged higher per unit prices. The initial consumption block can even be offered free of charge, provided that the prices pertaining to higher consumption blocks are sufficient to generate the necessary revenue in order to recover the costs. For example, in Johannesburg and Hong Kong, the first 6 m^3 of water consumed each month are provided at no charge.

In theory, increasing block-rate tariffs are designed to subsidise poor consumers who use small amounts of water per month compared to more affluent households that generally have higher water consumption (Whittington et al., 2009). However, this goal is not achieved in several cities in developing countries as many poor households in informal settlements are not yet connected to the water delivery system. Moreover, increasing the number of connections in poorer communities with lower water consumption will result in lower revenues for utilities, compared to increasing the number of connections to high-volume users. This notion may serve as a disincentive for the utility to increase access to the poor, especially in the case of profit-seeking private operators.

Additionally, IBTs face criticism for a number of reasons (e.g. Boland and Whittington, 1998; Hoque and Wichelns, 2013):

(i) all water consumers are subsidized regardless of their income (ii) the heavily subsidized first block is in most cases far too large due to miscalculations or political decisions; as a consequence the following blocks are not at all able to recover the costs and in some cases rich households actually receive more subsidies in total than the poor (iii) poor households often use a shared connection for many families and thus need more water in total from the second and following blocks than the average income family (iv) if poor households do not use a shared connection there is little to no incentive for water providers to connect them to the water supply network as they will actually reduce revenue.

4.3 Social Tariff Discounts

In Singapore, low-income households that are having difficulties in paying their water bills, receive assistance in the form of Utilities-Save (U-Save) rebates. This involves placing a fixed amount of cash in the household's utility account, which can be drawn from at any time to pay any utility bill, including water.

In London, customers defined as 'vulnerable' by the government may reduce their bills under the WaterSure scheme, by paying GBP 333 per year (GBP 199 for water and GBP 134 for wastewater) or their metered bill (if it is lower).

Eligible customers are those on certain specified state benefits who either have large families or use significant additional volumes of water because someone residing in the household has a qualifying medical condition (Thames Water, 2012).

4.4 Demand Managing Tariffs

In 1993, the Los Angeles Department of Water and Power (LADWP) restructured its water rates to provide customers with a clear financial signal to use water efficiently. The LADWP introduced a two-tiered, seasonal pricing structure, in which higher volumetric rates are charged during the high season (summer) compared to the low season (winter).

During periods of drought, shortage-year rates are implemented, during which the switch-point between the first and second tiers is reduced by 10 %, 15 % or 25 % to encourage additional water conservation and also to offset any revenue losses resulting from lower consumption.

This seasonally tiered tariff, along with other demand management strategies, has allowed Los Angeles to reduce its total water consumption continuously since 1990, despite the increase in population.

4.5 Subsidised Tariffs

The vast majority of countries in the world have a melange of subsidies (for socio-political reasons) and taxes (to generate public income and / or to drive water efficiency, demand management) incorporated in their water / wastewater tariffs. Subsidies have to

be raised through taxes, ODA or other sources and these are always limited. Water scarcity caused by demand greater than supply is in many countries as a consequence of subsidized tariffs that lead to an overuse of water resources (Zetland and Gasson, 2013). Moreover, underpricing of water often results in insufficient funds for the water utility and consequently in poor service and maintenance as was illustrated with the Vicious Circle of Water Services (see Figure 3 in Chapter 2). With water services and quality below a certain standard (e.g. intermittent supply, drinking water contamination) consumers need to invest in coping mechanisms (e.g. pumps, roof top tanks, water filters). Comparatively, these coping costs are much higher for poor households than for wealthier households since the investment needed is similar but the income level is very different. In conclusion, the poor might have to pay more money for their water than the rich despite the subsidies that are supposed to support them (Aggarwal et al., 2013).

It is generally assumed that subsidising tariffs is a problem that occurs in developing countries but it is true even for very wealthy countries like Saudi Arabia, as GWI (C. Gasson, 2013, p.5) has stated:

> "*Every year the government has reaffirmed its unwillingness to increase tariffs, even as the date of economic meltdown creeps closer and closer.*
>
> *This level of subsidy doesn't just take the lid off demand. It also undermines innovation in the Kingdom. Who is going to bother inventing a better mousetrap if the existing mousetrap is sold at 0.3 % of its actual value?*
>
> *It is no surprise that, despite spending $ 6 billion a year on water infrastructure, Saudi Arabia has not developed any domestic water technology companies of any note. There is just no incentive to invent low-energy processes or water-efficient systems if water and power are so cheap. ... The subsidy were to be applied in the form of a customer credit to the end user, then at least there would be an incentive for efficiency all down the value chain. ... If every cubic metre of water lost represented $ 10 leaking from the National Water Company's bank account, you can be sure that unaccounted-for water would be below 5 % within a short space of time. Plenty of local companies would see the opportunity to innovate with technology .... Even without asking consumers to change their habits, there is probably enough inefficiency in the system to meet rising demand for the next six years.*

It would be possible for the minister to commit to zero increases in production capacity, oil consumption and government funding by 2025, if the customer credit system could be used to manage demand among wealthy users as well. Who knows? We might also end up with a healthy domestic water and power technology sector to supply all the innovation required."

This statement illustrates the core problem of subsidized tariffs that very often cause more harm than good in the long run very well.

4.6 Water Tariffs and Non Revenue Water

All water tariffs are indirectly linked to non-revenue water. Suitable tariffs reduce non-revenue water while unsuitably designed tariffs are likely to increase non revenue water. The reason for this link of water tariffs with non revenue water will be illustrated in this chapter.

Zetland and Gasson (2013) conducted an extensive study using a data set on 308 cities in 102 countries in order to assess the relationship between water charges and sustainability as well as water efficiency. The found that subsidies and underpricing that kept water prices charges below the actual cost of service were apt to decrease water efficiency and thus to increase overall water demand. Moreover, higher water prices and good governance were correlated.

To determine a suitable water tariff that influences the amount of non revenue water the total costs of water supply have to be calculated. The best approach on how to calculate the actual costs and how to measure water consumption efficiently at low costs, especially for agriculture, is still a topic of debate. Nevertheless, the majority of studies published in regards to water pricing agree that charging water prices that meet operating and maintenance costs and as well as a part of the capital costs creates medium and long term benefits (e.g. Gawel and Fälsch, 2012; Nikouei und Ward, 2013; Zetland and Gasson, 2013). Yet – regardless of academic consensus – in nearly all emerging and developing countries cost recovery goals, if existing, are usually limited to recover O&M expenses and even this is often not achieved as most tariffs levels are far below actual costs (e.g. Molle and Berkoff, 2007; UNDP, 2006).

On a medium and long-term perspective, the economic losses and the thereby created lack of funds for operation, maintenance and investing can be identified as a major factor contributing to physical water losses. Hence the amount of non revenue water increases.

The rationale for this is that the technical requirements to repair and maintain basic water infrastructure and thus for water loss reduction have not changed substantially within the last decades and could in most cases be carried out by the respective local organisations if there were sufficient financial resources available.
Accordingly, there is a positive correlation of cost-covering tariffs (including investment costs) and the reduction of water loss rates.

The capital city of the Philippines, Manila and its metropolitan area serves as an example of how an improved tariff system can reduce water losses. Since 1997, the responsibilities for water and wastewater services in Metro Manila have been delegated to Manila Water Company, Inc. (MWCI) for the east zone (1400 km²) and to Maynilad Water Services, Inc. (MWSI) for the west zone (540km²) via two 25-year concessions, which were extended by another 15 years in 2009.

MWCI has significantly reduced its NRW, from 63 % in 1997 to 11.2 % in 2011, and improved its reliability, in terms of 24-hour availability, from 26 % of customers in 1997 to 99 % in 2011 (Manila Water, 2011).

MWCI implements a complex six-component tariff structure. For residential consumers, this consists of a basic charge of PHP 101.01 (USD 2.33) per connection for the first 10m³ of water, an IBT consisting of 9 tiers, a foreign currency differential adjustment of 1.23 % of the water charge, an environmental charge of 20 %, a sewerage charge of zero (compared to 30 % for business groups 1 and 2), and a maintenance service charge of PHP 1.50 - 50 per connection for a range of meter sizes (Manila Water, 2012).
MWCI has been implementing a social lifeline tariff rate for low-income consumers using less than 10 m³ per month, and in 2012, while tariffs for all other categories increased, this social rate was reduced further, such that low-income consumers (accounting for about 8 % of all consumers) now enjoy a 40 % discount.

5. THE IMPACT OF WATER PRICING ON DIFFERENT USER GROUPS

In order to prevent river basin closure IWRM is crucial and as was shown in the previous chapters, one of the key aspects of IWRM is pricing water according to its value and with consideration of social aspects to improve water use efficiency.

The previous chapter illustrated how different tariff structures can, in general, affect water consumption. This chapter will provide a more detailed analysis of the impact on the three user groups in the project region (Agriculture, Industry, and Domestic).

Water efficiency in Iran and most developing and emerging countries is fairly low. Major factors that cause low water efficiency are high water losses due to leakage, waste of water resources due to high subsidies and water charges unrelated to actual consumption (e. g. flat rate tariffs). To determine the value of water for different users while considering social aspects and implications will help to improve water efficiency and to determine how to allocate the scarce and depleting water resources available in the Zayandeh Rud Catchment Area between all users and the environment. The basic arguments for pricing irrigation are comparable to the arguments for water charges in general. Water charges serve many purposes (cost recovery, sustainability, equity, allocation) as summarized in Figure 6 and shown in Chapter 4.

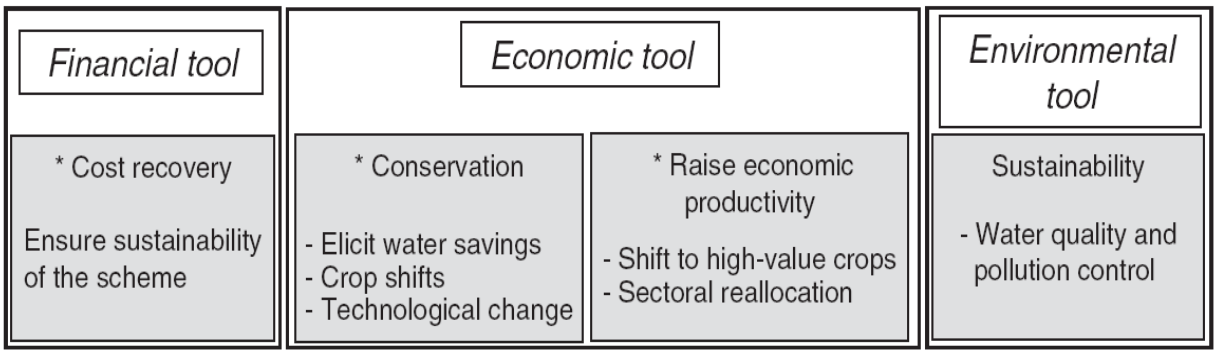


Figure 6: Different purposes of water charges
[Source: Molle and Berkoff, 2007, p. 22]

5.1 Agriculture

This chapter focuses on water pricing for irrigation as the agricultural sector accounts for most of the water withdrawals in the Zayandeh Rud Catchment Area. Water pricing will affect the water consumption and allocation within the river basin. Yet the effects of water pricing are difficult to forecast.

In 2004 the agricultural sector accounted for more than 90 % of the total water withdrawal in Iran (FAO, 2013); hence improving water efficiency in irrigation is seen as a key component to reduce water pressure in Iran. Especially, as it plays a vital part in securing food supply for a fast growing population and economy (e.g. Nikouei und Ward, 2013; Ward and Pulido-Velazquez, 2008; Fang et al., 2007).

Improved irrigation offers many benefits as well. These benefits include increased yields, reduced water charges and improved quality of the agricultural products. The extent of these benefits depends on numerous factors such as the size and type of farm, the costs of improved irrigation and the level of water charges (Venot and Molle, 2008). Moreover, and in contrast to Molle and Berkoff (2007), Panahi et al. (2009) state that in Iran, as elsewhere, agricultural water consumption is inefficient but could be lowered substantially by using drip and micro-irrigation.

Molle and Berkoff (2007) argue, however, that water efficiency in irrigation is higher than generally assumed and in the case of irrigation pricing the above mentioned purposes such as economic efficiency are rarely achieved. In their view this is due to flawed cost-benefit analysis that tends to disregard many costs and is biased towards new construction as well as decisions that are often based on (geo-) political objectives and disregard the challenges of shifting to high-value crops (lack of funds, expertise etc.). In addition, farmers should not be forced to pay for the capital costs of oversized or inefficient infrastructure that was not build according to their needs but to political decisions. The arguments presented by Molle and Berkoff on water efficiency in irrigation are, however, mainly based on anecdotal evidence. This report will therefore follow the consensus of the majority of scholars, especially as irrigation in Iran is mostly based on flood irrigation.

In the case of Iran and the Zayandeh Rud Catchment Area in particular, the system of water supply by irrigation ditches and flood irrigation (border dyke irrigation) prevails.

This system has proved to be reliable and relatively inexpensive; yet a multiple of the water quantities are needed which would be consumed with more efficient systems and technologies (e.g. drip irrigation). With heavily subsidized tariffs farmers do not have any incentive to invest in more efficient irrigation techniques as for them the best business decision is to water the plants for as little money as possible which in this case is flood irrigation. Once the prices for water are raised investments in more efficient irrigation start to make sense financially. Moreover, this would reduce the amount of Iranian water indirectly exported to other countries as a lot of water is required to produce agricultural products.

Governments often promote and subsidise the use of drip irrigation and similar systems yet there are a number of potential difficulties that need to be accounted for. First, in the case of Iran exceptionally high temperatures in summer of up to 40 °C and above cause very high evaporation. Second, subsidies might not be distributed evenly (e.g. lack of awareness from famers, lack of will or funds to afford the technology even though it is already subsidised, lack of expertise) which would create a competitive advantage for farmers that are able to receive the subsidies. Third, findings by Ward and Pulido-Velazquez (2008) show that in North America with higher subsidies on drip irrigation farmers often use the water saved, to irrigate an increased area of farmland, which is especially true for countries were land is relatively cheap and/or abundant (Molle et al., 2010). Fourth, more efficient irrigation may reduce return flow and aquifer recharge that is of great importance to river basins. Fifth, farmers that face higher water charges are likely start/increase the (illegal) withdrawal of groundwater for irrigation purposes. Due to hydrological interdependence all interactions such as groundwater tapping, changes in water withdrawals and land use in the river basin will have up- and downstream impacts (Molle and Berkoff, 2007, Molle et al., 2010).

Jordan, as one of the countries with a similar climate to the mid of Iran, very scarce water resources and agriculture as the main water user as well, offers a valuable insight on water pricing policies and their effects on irrigation with a focus on groundwater depletion. In Jordan, water pricing policies were introduced that aimed to reduce groundwater discharge by agricultural users up to the natural recharge rate by increasing volumetric charges. The efforts were, however, not successful for a number of reasons including (i) metering of withdrawn groundwater proved to be very problematic; (ii) water prices were not raised high enough / the quantity that was allowed to be abstracted free of charge was too high; (iii) prices at this level decreased the revenue of

low income farmers with low-value or extensive crop that mostly were not able to successfully invest in measures to improve efficiency and output; (iv) the anticipated shift towards high value crops took not place as water prices were either still too low or farmer lacked the means or will to change their crops (Venot and Molle, 2008). For water pricing in the Zayandeh Rud catchment area the here discussed conclusions as well as the results of similar cases studies need to be taken into account in order to establish a price system for agriculture suitable for this particular area.

In Iran, non-local irrigation ditches have mainly been installed by public/government funds and therefore often lack any sustainable management and maintenance by local or private bodies. Other local irrigation ditches that serve the purpose of distributing the water evenly on the agricultural land and connecting agricultural areas close to each other were built by organised interest groups of farmers or individual farms and are also managed by them.

To face some of the above mentioned challenges in irrigation water pricing Nikouei and Ward (2013) conducted an extensive study using a water balance and crop use model to assess the existing BACP (below average cost pricing) model in Iran. Based on the findings of their study they promote a two tiered pricing system in the Zayandeh Rud River Basin in central Iran that in contrast to the existing BACP has prices slightly above marginal costs in times of sufficient water supply that increase during times of drought for discretionary use of water. As it is common in two or more tiered pricing systems, the price of water in the first block should be very low with a quantity that covers the minimum water requirement of one household and cross-subsidised by the more expensive prices in higher tiers. This way farms will ideally always be able to secure their own food supply with affordable water prices for irrigation. The more expensive water prices in the second tier, especially in times of water scarcity, will promote water efficiency. The study by Nikouei and Ward (2013) concludes that the two tiered pricing system would reduce prices for securing the minimum level of water supply for subsistence use while charging higher prices for discretionary uses and thus promote water efficiency.

A study on the institutional mechanisms that would offer farmers an incentive to reduce water consumption by water conservation measures conducted by Nikouei et al. (2012) in the Zayandeh Rud River Basin revealed that a water conservation subsidy (i.e. public financial support to implement water saving measures) would provide sufficient

incentive for farmers in Iran to introduce water saving technologies such as drip irrigation instead of flood irrigation. In theory this policy would increase crop yields as well the shadow price of water and the return on investment for the farmers. This result is very promising especially from an environmental perspective and considering the competition for water supply.

While farmers in Iran theoretically have to pay for the capital as well as O&M costs of extracting groundwater with the current pricing system, in reality there is no price charged for the amount of water extracted for irrigation purposes. As no incentives are offered for water efficiency groundwater resources are often overexploited (Nikouei und Ward, 2013). Furthermore, our own research clearly showed that the prices paid for water vary not only between users groups but also within user groups for no specific reason and in general prices are not sufficient to meet O&M expenses.

5.2 Industry

The industry, together with the domestic users, accounts only for a relatively small share of the total water demand but this share is crucial for the creation of value as well since the value added for industrial products is higher than for agricultural products and the Zayandeh Rud Region is the second largest industrial area in Iran; for example 50 % of steel produced in Iran is produced by steel mills in the Zayandeh Rud Region. Other main industries located in the Zayandeh Rud Basin are cement factories, oil refineries, the defence industry and a polyacrylic factory (Murray-Rust and Droogers, 2004). All together, the share of water consumption by the industrial sector in Iran is increasing (Rajabi, 2013) and is said to double within the next ten years.

The cumulative water demand for industrial water consumption has increased sharply since 1974 (Murray-Rust and Droogers, 2004). The water efficiency is, however, low. A study conducted by Rajabi (2013) that showed the impact of water pricing on the industrial sector in Isfahan, concluded that it can be expected that increasing water prices will increase the water efficiency significantly. This means that the industry is able to reduce water consumption if there is a monetary incentive to save water.

Unfortunately, the exact water demand for the Industry in the Zayandeh Rud River Basin could not be determined as there was no reliable data available.

From the data and discussions with officials that were held, there have been indications that the demand is rising – which are consistent with the overall water consumption of the industrial sector in Iran.

5.3 Domestic

The water demand of private households is – compared to the agricultural – small, yet considering the social aspects of water supply, private households are at least equally as important. Expected changes once the water prices are increased are, apart from decreased water wasting per se – in Germany the average consumption of water lies at 121 litres per person per day (DESTATIS, 2014) compared to a consumption of around 250 litres per day per person in urban areas – is the introduction of water efficient household appliances. However, this will not be possible for all households since the money for the initial investment will not be readily available and neither will all people be convinced that the appliances will actually save them money in the long run.

Furthermore, some households in the target region are not using any or only very few household appliances yet. Consequently, an increase in price might not have a significant impact of the use of water saving technologies within private households. It will however reduce overall water consumption by private households as was for example shown in studies conducted for Europe (EEA, 2013) and worldwide (Zetland and Gasson, 2013).

As of now the water price charged in Iran covers only around 10 % of the total costs and thus does not provide any incentive for households to invest in water saving technologies (Yousefi, 2010).

5.4 Conclusion

In conclusion, the impact of water pricing on different user groups differs and is likely related to the specific value of water as well as the type of water supply. These differences and challenges that go along with setting suitable water tariffs in the context of IWRM need to be acknowledged and taken into account if a functioning IWRM is to be achieved. There is no "one-size-fits-all" approach for IWRM. The basic approach always needs to be adapted significantly to the conditions in the target area. In the case of Iran and the Zayandeh Rud Catchment area in particular, the pricing impacts on the

various sectors can be summarised as follows. For the agricultural sector the effects of water pricing are hard to predict. As of now water efficiency is very low but has the potential to be increased significantly and relatively simple with the introduction of improved irrigation techniques. The studies that were analysed in this report showed that higher water prices and conservation subsidies would provide the necessary incentives to water saving technologies. A successful implementation would likely improve crop yields, the shadow price of water and the return on investment for the farmers. However, there are several challenges that are to be faced (e.g. how to meter groundwater withdrawals; how to calculate new water prices in a precise and fair manner; how to avoid the fact that higher prices decrease the revenue of low income farmers with low-value crop that are often not able to invest in measures to improve efficiency).

The water demand of industrial users in the Zayandeh Rud Basin has increased significantly within the last decades. The creation of value in the industrial sector is much higher than in the agricultural sector (see Chapter 7 & 8) even though the water efficiency is poor. A study by Rajabi (2013) for the industrial sector in Isfahan stated that a raise in water prices will increase water efficiency within the industry significantly.

The demand of the domestic users is – while low in contrast to the agricultural demand – very high if one compares the consumption per day per capita to most industrialised countries. Water tariffs for domestic uses are a crucial aspect of IWRM as water is not only an economic but also social good. The impact of increased prices on household consumption have been a topic of debate for many years and most studies conclude that increased prices will reduce the consumption yet tariffs need to remain fair. In contrast to the old paradigm of heavily subsidised tariffs (water tariffs in Iran only covers around 10 % of the actual costs) it is nowadays increasingly stated to stop subsidizing the tariffs and to start subsidizing the poor. The reasons for this rationale were analysed in the previous chapter.

6. INFORMATION BASE AND METHODOLOGICAL APPROACH

The collection of data and analysis of the technical, organizational, economic and environmental and water management situation in the catchment area was conducted throughout the project duration. A special focus was placed on the different water users (domestic, agriculture, industry) and the possible impact of different water prices on their demand as well as to determining their individual value of water. Unfortunately, the figures provided by the Iranian side contradicted each other frequently, resulting in a situation where a scientific evaluation and a meaningful interpretation of data was significantly more difficult than usual or in many cases not possible at all. The following table created by the authors for the price of water per cbm for domestic users in 2002 based on data provided by the Iranian side exemplarily illustrates this problematic.

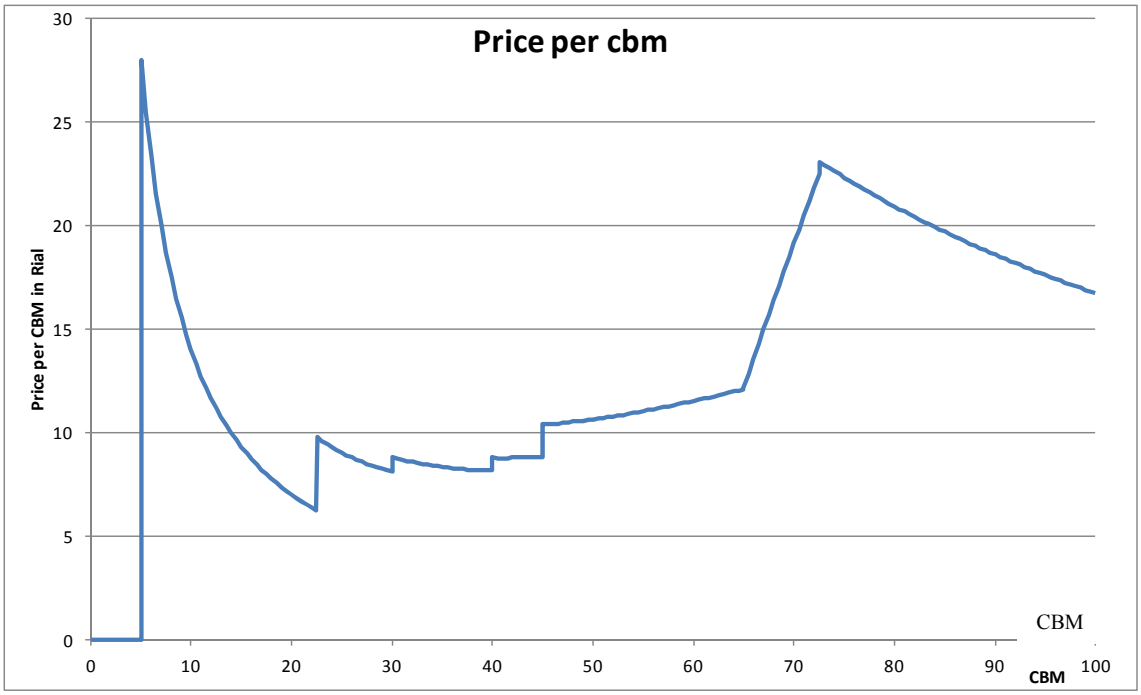


Figure 7: Price per cbm for potable water
[Source: IEEM 2013

This unusual curve progression may in parts be explained by the fact that up to 2010 water tariffs were to some extend based on consumption patterns that are no longer available and therefore not comprehensible.

This variety of prices that does not seem to follow a particular logic may be caused by the fact that Iran is a "budget economy" and there are very few, if any, real prices, which have arisen on a free market basis show the value of the buyer / seller for goods and services. Furthermore, economic data are not collected regularly nor are they collected in a standardized form. In addition, a disclosure of the data is usually subject to the approval of several decision-makers from the Iranian side, which in turn meant that not many data sets were submitted and if they were, they often were incomplete or incorrect.

Basically, what could be concluded from the data is that the conditions in the target area, as was to be expected, differ greatly from the conditions in developed countries, (e.g. Germany); but they also differ slightly from other developing countries, inter alia, by having a public infrastructure that does not include the complete services of water supply. Depending on the individual case, the user group (domestic users, agriculture, industry) and the hydrological conditions (supply constraints permanently or only during dry periods), a water infrastructure has been developed in which the personal contributions of the users account for a relatively high share of the water supply system.

In agriculture, the system of water distribution through irrigation ditches and flood irrigation (border dyke irrigation) dominates. This system is robust and comparatively inexpensive; but a multiple of the amount of water needed, which would be consumed for more efficient systems and technologies. The non-local irrigation ditches were mostly funded by public institutions/authorities and are supposed to be maintained by them as well yet a sustainable management and sufficient maintenance is often not achieved. The irrigation ditches for distributing the water on the agricultural land or between individual agricultural lands were, however, built by numerous differently organized farmers or individual farms and are managed by them privately.

Similarly, while the water user group industry is supplied by publicly funded water infrastructure when, for example, small businesses or the local industry are directly connected to the public water supply infrastructure within the city, in most cases, the industry is supplied by superregional water supply systems and has to provide substantial personal investments, e.g. for water storage, filtration etc. In many cases, the industry supplies itself independently.

Domestic users are served to tap, with a certain pressure and no own infrastructure to be built and maintained, except the in-house installations and (maybe) some post-filtering, disinfection, storage (e. g. roof tanks).

The basic data needed for the construction of a curve for supply and demand applicable for IWRM (as explained in general with Figure 4) would be the water tariff set, the water revenues collected in the target region, differentiated for different consumer groups (like domestic consumers, agricultural irrigation, industrial users etc.), in relation to the water consumption (cubic meter per capita and year or else). Such data has in parts been sent to the authors and analysed accordingly. Unfortunately, the data did not seem to be consistent, the water tariffs reported did not match with the revenues reported, and there were great differences between different reporting years and non-plausible differences between neighbouring utilities etc.

After considering different approaches (including the ex-post completion of tariff data from the target area, which turned out to be impossible), the research team decided to use representative and proven data from other water service areas, which seem to be comparable and representative for the target region, the Zayandeh Rud Basin.

Given the observed variation in tariff components and water prices across cities (OECD, GWI, IWA, IB-Net, WB et al.), generalized conclusions at the global or regional level could be misleading. However, a snapshot of the domestic water and wastewater bills from the present sample reveals that average unit bills in Asia and Africa are generally less than USD 1.00/m³, which is far below average for all of the cities considered (USD 2.10/m³). In Europe, North America and Australia, the unit bill is generally higher than USD 3.50/m³. The GWI (2012) survey of domestic water and wastewater tariffs in 310 cities showed similar results. Of the 51 cities with a unit bill greater than USD 4.00/m³, 39 are in Western Europe, 6 in North America, 5 in Australia, and 1 in Latin America Similarly, of the 81 cities with a unit bill less than USD 0.50/m³, 49 are in South, East or South-East Asia, while 13 are in the Middle East, North Africa, or Sub-Saharan Africa (GWI, 2012).

On average, households spend around 1.5 % of their monthly income on their water and wastewater bills. In a few European, Australian and American cities households spend more than 2 % and in several Asian cities, although monthly bills are quite low compared to the absolute amount, the water bill represents a notable part of the incomes

earned by poorer households in those cities. The average household income for each city was used when performing these calculations. Yet, incomes vary substantially, particularly in developing cities, so that the actual proportions for many residents can be larger than the average values reported here.

In a first step, the average tariffs (USD/m³) and water consumption rates (L/capita/day) were listed for in total 192 cities from 19 selected major countries in the years 2008 to 2012.

Figure 8 for 2012 and (to compare with 4 years earlier) 2008 visualises, what consumers do pay at what rates of water consumption, and the same for water plus wastewater services (in total). The regression curve is presented in the figures. However, under the presumption that only the higher water tariffs are cost-covering, and subsidised tariffs lead to a certain consumer's profit, the upper curve including higher values are the ones, which can be taken as "reality" of what consumers are willing to pay.

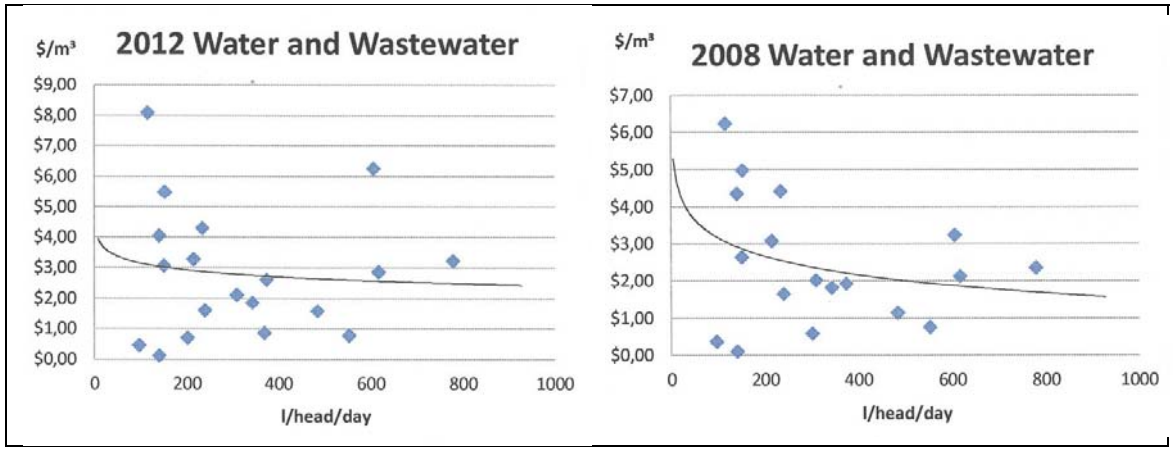


Figure 8: Average Tariffs (USD/m³) and Water usage in selected major countries in 2012 and 2008 [Rudolph, Kluska, Strömer, 2013, basing on GWI Data; see Attachment 10]

7. THE VALUE OF WATER SERVICES

Comparing the city of Isfahan with around 70 cities, for which data files have been collected, the city of Pretoria in South Africa seems to be quite similar in terms of water resources availability (Isfahan has even less precipitation mm p.a. and fewer precipitation days on average, while both are relatively arid cities), the per capita income and water related infrastructure as well as the settlement structure – South Africa's GDP (PPP) is generally very similar to the GDP of Iran. The per capita consumption (as quite often) varies substantially for different houses, families, locations within the supply area (and, of course, very different depending from the continuity of water supply, supply pressure etc.). The figures in Isfahan as well as in Pretoria are said to range between 30 and 450 litres per capita and day (taking into account only those consumers which are connected to the reticulation network, not taking into account customers served through water trucks or stand pipes).

According to this consumption, the value of water per cubic meter would range between 3 and 8 USD/m³. In average over the whole service area, one could well count with 5 USD/m³.

This amount looks very high, compared to the collected water revenue, which is well below 0.40 USD/m³. It does not look high, compared to the efforts and public spending from the state budget to build the water infrastructure, once these expenses are divided through the cubic meters of water sold (not through the cubic meter of water that could be produced and distributed according to the technical capacity, which is about 2 times higher).

As discussed in Chapter 6, domestic consumers are served to tap, with little or no own infrastructure to be built and maintained, apart from possible in-house installations and in rare case post-filtering, disinfection, storage (e.g. roof tanks) etc.

This situation is different from the agricultural consumers, operating their own border dyke systems (flood irrigation) respectively pistol or pyrite irrigators, even drip irrigation. Agricultural consumers even run own pumping stations to abstract irrigation water from the rivers.

Therefore, the value of agricultural water supplied is significantly below the value of water supplied to domestic consumers (acknowledging that wastewater is utilised or done at negligible expenses. Taking data from other IWRM driven surveys and the information and results of discussions held with experts in the target region, the figures are the following:

Table1: Estimated Water Values in Agriculture for Water Supply Services

	% of Domestic Supply	Value USD/m³
Flood Irrigation	2	0.10
Conventional Irrigators	4	0.20
Drip Irrigation	15	0.75
High Value Crops (orchards, vegetable etc.)	30	1.50
Life stock	40	2.00
Sum respectively Average (roughly estimated)		
Low Water Efficiency	4	0.20
Medium Water Efficiency	8	0.40
High Water Efficiency	20	1.00

For industry it is hard to estimate the value of water, due to the fact that there are no data about the kind and productivity of industries consuming water in the target region. Cooling water abstracted by industry itself is excluded from the valuation (only supplied water and wastewater services shall be valued, managed under public governance under IWRM). The following average analysed from other countries can be quoted.

Table2: Water Value for industrial supply at the current level of water efficiency

	% of Domestic Supply	Value USD/m³
Small Industries in Town	150	7.50
Wealthy & High Tech Industries	150	7.50
Conventional Branches	50	2.50
Mining & Agricultural Processing	30	1.50
Average (rough estimation)		4.00

8. OUTLINE OF WATER VALUES FOR THREE SCENARIO DEVELOPMENTS

Based on the scenario developments to be considered in the project, as developed through project partner ISOE (see presentation "Water Management in the Zayandeh Rud Basin: Three Scenarios by E. Schramm and E. S. Sattary, 2013) IWRM shall be elaborated in three perspectives which are

- a trend scenario
  (status quo forecast with "business as usual"),
- a modified outlook scenario
  (some IWRM coordination primarily in Isfahan province) and
- a maximum scenario
  (full IWRM including the second province Gaw-Khuni).

Whatever shall be decided in concrete actions and details regarding these three scenarios, it is quit obvious that major parameters influencing the economic value of water are related to these three scenarios, namely

- the productivity of water utilisation
  (i. e. the allocation of available water resources: providing water for those purposes which generate greater benefit),
- the coverage of water supply and sanitation
  (in location as well as in time: more consumers served respectively increasing coverage; improved continuity of services without regular interruptions during peak-hours or dry seasons).

It is clear that advanced IWRM measures lead to improved conditions in water efficiency and water effectiveness, as described above. On the other hand, there will be increased financial and political efforts needed to reach the advanced status of water services and IWRM. (In this chapter, only the value of water is dealt with; the costs of water services are examined in other chapters of the research project.)

Related to the description of the three IWRM scenarios, three levels of water efficiency respectively water service affectivity (which might be measured with the indicators for water and sanitation service performance well known from the literature, see World Bank toolkit; see Rudolph 2013). The following table indicates these figures

Table 3: Value of full water supply and wastewater sanitation services

Consumer Sector	Low Water Efficiency (trend scenario)	Medium Water Efficiency (modified scenario)	High Water Efficiency (maximum scenario)
Settlements (domestic consumers)	3.0 USD/m³	5.0 USD/m³	8.0 USD/m³
Agriculture (mainly for irrigation and livestock)	0.2 USD/m³	0.4 USD/m³	1.2 USD/m³
Industry (productivity and profitability increasing with water efficiency)	4.0 USD/m³	7.0 USD/m³	14.0 USD/m³

The figures above indicate the value of the full range of water supply and wastewater management (including disposal as well as reuse). However, especially in agriculture and also in industry, it has to be understood that the public utilities do not provide water services for all technical stages. Therefore, it might be necessary to separate the full value of water along the supply chain (technical measures), for which an estimation is listed below.

Table 4: Distribution of full values along the value chain of water supply and wastewater management

Water Supply	% of Full Service Value	Wastewater Management	% of Full Service Value
Raw Water Provision	20	WW Collection (tertiary and secondary sewerage)	35
(Bulk Water Provision incl. Purification)	30	WW Transportation (primary sewerage)	20
Delivery of Water to User (all services including technical as well as managerial / administrative activities. Storage, metering, billing and collection included in one of the activities mentioned above, depending on the local conditions)	50	Stormwater Collection	15
		Wastewater Treatment (stormwater treatment included)	30

Therefore, in case that a farm does its own local water distribution (e. g. through border dykes or drip irrigation), these components of the supply value chain are not to be counted on the utility's invoice, but deducted from the value of water services provided through the utility under discussion. In this case, the water utility would account for 100 - 50 = 50 % of the total water value generated.

9. CONCLUSION AND OUTLOOK

This study was conducted to determine the value of water for different users groups (Domestic, Industry and Agriculture). This, however, proved to be very challenging for a number of reasons: (i) a lot of the raw data was provided in Farsi and had to be translated which caused a delay as well as the possibility of imprecision (Attachment 8 serves as an example) (ii) the data basis was political prices (budget economy) (iii) in many cases the data did not add up due to simple miscalculations or contradicting statements (e.g. the water tariffs reported did not match with the revenues reported, and water prices varied substantially for water users of the same group) (iv) the data that was obtained and made sense was not sufficient to produce viable conclusions (v) disclosure of the data is usually subject to the approval of several decision-makers from the Iranian side, which in turn meant that not many data sets were submitted and if they were, they often were incomplete and or incorrect.

These challenges led to many in-depth discussions with Iranian and German experts and offered new perspectives on the Iranian water sector but as a consequence and after considering various approaches (including the ex-post completion of tariff data from the target area, which turned out to be impossible) the decision had to be made to base the study on representative and proven data from other water service areas, which were comparable and representative for the target region, the Zayandeh Rud Basin. This decision was made in consultation with and agreement of the project partners and the Project Management Agency Karlsruhe Institute of Technology (PTKA-WTE). This decision enabled the authors of this report to determine a value of water for all three scenarios and water users that should be close to reality and hence fulfil the research objective as far as was possible.

In a first step, the average tariffs (USD/m³) and water consumption rates (L/capita/day) were listed for in total 192 cities from 19 selected major countries in the years 2008 to 2012. In a second step the city of Isfahan was compared in more detail with around 70 cities, for which data files have been collected and analysed. The city of Pretoria in South Africa was found to be quite similar in terms of water resources availability (both are relatively arid cities), the per capita income and water related infrastructure as well as the settlement structure and thus taken as model.

The results on the value of water services for Domestic, Industry and Agriculture user can be summarised as follows: The value of agricultural water supplied is significantly below the value of water supplied to domestic consumers. The value of water for industrial users is slightly higher than for domestic users. In all cases the value of water is significantly higher than the price of water.

In determining the value of water for different users it has to be understood that the public utilities do not provide water services for all technical stages. Hence, the value actually provided by the water utility will be lower than the full value of the water service since personal/private contributions were made to receive the water (in a sufficient quality).

Despite the findings on the value of water, impacts of pricing on the consumers as well as consequences for a sustainable water allocation in the target region, implementing a new tariff and water allocation structure that actually reflects the value of water to some extent will be very challenging in budget economy such as Iran and should be seen as a political process as well. This finding is nicely reflected by the following statement from Nikouei and Ward (2013, p. 43)

> *Translating our results into a just water rights administration system will be difficult in Iran or other developing countries with world views in rural areas different from policy analysts schooled in western analytical thinking. That is, policy insights gained by rigorous scientific analysis of objective data have no guarantee of being used to inform or support policy.*

All in all, the current water pricing and allocation system is clearly failing and has to change as demand is increasing while supply and quality are declining simultaneously, leading to an intensified competition for water between and within the user groups. This situation might be able to create enough pressure on the authorities for a change of paradigm of water management in Iran.

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11. ATTACHMENTS

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Berichtsblatt

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18. Kurzfassung:
Die Lage im Projektgebiet – das Zayandeh Rud Einzugsgebiet – ist geprägt von Wasserknappheit, sinkender Wasserqualität und der drohenden Zerstörung des Ökosystems vom Salzsee Gavkhouni. Ein Grund dafür sind neben dem steigenden Wasserbedarf und der Ableitung von ungeklärten Abwässern, die sehr niedrigen Wassertarife, die auf Basis politischer Entscheidungen festgelegt werden und keine Allokation des oder Einsparung des Rohstoffes Wasser bewirken. IEEM unterlag im Rahmen des Verbundprojektes die Durchführung des Unterprojektes C: „Wert des Wassers“. Ziel war die Bestimmung des Werts des Wassers für die verschiedenen Nutzergruppen (Landwirtschaft, Industrie und Haushalte) im Zayandeh Rud Einzugsgebiet. Hierzu war die Erfassung und Analyse der technischen, organisatorischen, ökonomischen und ökologischen wasserwirtschaftlichen Ist-Situation im Einzugsgebiet erforderlich, bei der ein spezieller Fokus auf die separate Betrachtung der unterschiedlichen Wassernutzer gelegt wurde. Diverse Datensätze und Informationen wurden übersetzt und ausgewertet. Die von der iranischen Seite übermittelten Zahlen widersprachen sich vielfach und waren nicht vollständig, so dass eine gezielte Auswertung und eine sinnvolle Interpretation maßgeblich erschwert wurden bzw. nicht möglich waren. Dies mag zum einen an der Tatsache liegen, dass der Iran eine "Budgetökonomie" ist und es kaum echte Preise gibt, welche am freien Markt entstehen. Zum anderen werden ökonomische Daten nicht in der erforderlichen Form erhoben und weitergeleitet. Aufgrund der voranstehend genannten Probleme hat sich das Team des IEEM darauf verständigt, eine Vergleichsanalyse durchzuführen, welche zum Ziel hatte, die Zahlungsbereitschaft der verschiedenen Nutzergruppen (und somit deren individuellen "Wert des Wassers") aus den Erfahrungen in anderen Ländern mit ähnlichen Rahmenbedingungen abzuleiten unter Berücksichtigung der drei für das Projekt entwickelten Szenarien und somit das Projektziel soweit wie möglich zu erfüllen. Die Ergebnisse zum Wert des Wassers können wie folgt zusammengefasst werden: der Wert des Wassers für landwirtschaftliche Nutzung ist deutlich geringer als der Wert des Wassers für private Haushalte deren Wert wiederum unter dem der industriellen Nutzung von Wasser liegt.

19. Schlagwörter
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18. abstract
The project's objective was to conduct research on the economic value of water in the Zayandeh Rud Catchment area for different scenarios and users. As of now setting water tariffs in Iran is a political decision that does not reflect the value of water or the costs of supplying it. The tariffs are low and the water sources in the Zayandeh Rud Basin are overused - water scarcity, poor water quality and the imminent drying up of the important Gavkhouni swamp are the consequences. The combined effects of increased water demand and reduced water supply have caused intensified competition among the different water user groups (i. e. agriculture, industry and domestic users). It is thus not possible to simultaneously fulfil the equity, efficiency and sustainability objectives. Integrated water resources management (IWRM) is a suitable approach to improve the situation as it aims to manage and allocate water to create the maximum economic and welfare possible while preserving the ecosystem (GWP, 2000), in order to achieve that the value of water needs to be determined. This, however, proved to be very challenging for a number of reasons: (i) a lot of the raw data was provided in Farsi which caused delays (ii) the data was based on political prices (budget economy) and did not reflect the costs of providing water (iii) in many cases the data did not add up due to simple miscalculations or contradicting statements (iv) the data that was obtained and correct was not sufficient to produce viable conclusions. After considering different approaches (including the ex-post completion of tariff data from the target area, which turned out to be impossible), the research team decided to use representative and proven data from other water service areas to determine a value of water for three scenarios and all water users that should be close to reality and hence fulfil the research objective as far as was possible. The results can be summarised as follows: The value of agricultural water supplied is significantly below the value of water supplied to domestic consumers which in turn is slightly lower than for industrial users. In all cases the value of water is significantly higher than the price of water.

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